

Sasy 60i



Productcatalogue 2015

EATON

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Powering Business Worldwide

Customer-focused and Innovative SASY 60i - now a UL-certified Component

SASY 60i - safe and reliable: In combination with the new generation of Eaton's motor protectors and circuit breakers, SASY 60i provides a universal UL-certified solution for switching, protecting and distributing power.

The modular SASY 60i busbar system by Eaton has been conceived for the efficient distribution of power in the switching cabinet. Thanks to busbar adapters, feed and output switches can be mounted directly onto the busbar system in a quick and a space-saving way.



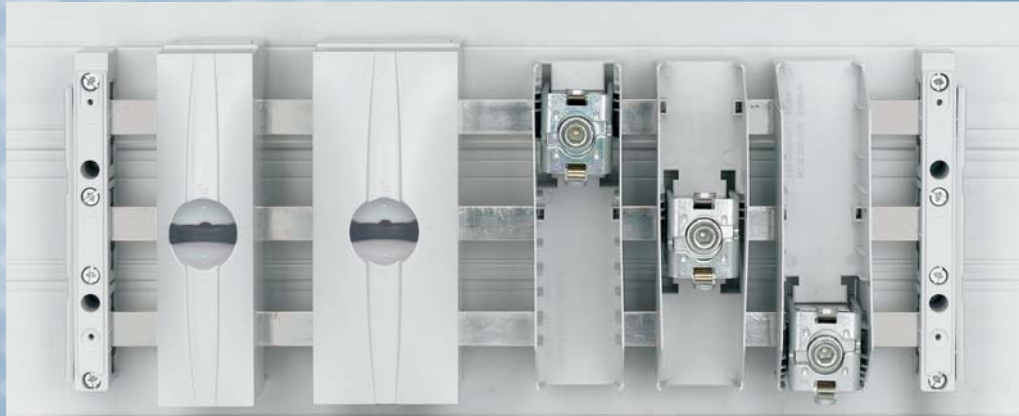
Optimized busbar profile

The system offers many advantages. For example, the SASY 60i uses double-T-profile bars, thus reducing the time and effort needed to prepare the contact points. The profile uses very few busbar supports for very high rated peak withstand currents (I_{pk}); it thus optimally utilizes the limited cabinet volume. In addition, dissipated heat is conducted in the best possible manner thanks to the large surface area of the busbar profile. Thanks to the market-conforming 60 mm center-to-center distance between the busbars, the system is compatible to other set-up components such as bus-mounting fuse bases or NH fuse switch disconnectors or the new D0-load disconnector switch using the D02-LTS/63/3-R type of fuse.

The latter provides for all-pole switching of the load (quick break) independent from manual switching, and safe fuse replacement in a voltage-free status. The device comes as standard with a flashing signal and contact position indicator, which either inform the user about a faulty fuse or show the switching status 0 or I of the device. The plug-type technology without fuse carriers (fuse plug) not only reduces the dissipated heat of the protective device while it is in operation, but also enables the user to replace a hot fuse after tripping without having to touch it with his hands. D02-LTS/63/3-R is available as a 3-pole and 4-pole version and it is extremely space-saving thanks to its overall width of 27 mm only. Retention springs making it easy to insert type D01 and cylindrical size 10x38 fuses in the fuse plugs are included in the scope of delivery. The load disconnector switch can of course be locked out and sealed.



The Busbar System for the Global Market



Short installation time thanks to pre-assembly

Eaton offers direct and reversing starters up to 15 kW, fully mounted on busbar adapters. These fully assembled units consist of one PKZM0 motor protector and one or two DILM contactor(s). In order to mount these, they only need to be clicked in place on the busbar; this guarantees reduced assembly times and costs.

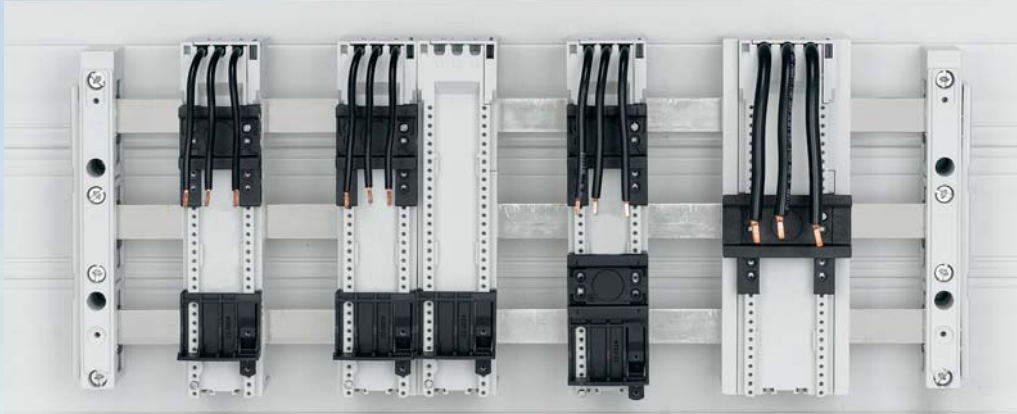
Special features of the device adapters

The device adapters offer a special functionality in that they can be mounted onto different profiles and busbar thicknesses. The adapters connect to the motor protector and circuit breaker directly over the busbars, comfortably and without requiring any boreholes, up to 630 A.

By reducing the width of the adapter to 45 mm, it has been possible to match it to the width of the motor protectors and contactors. The actual mounting surface on the busbar system is thus optimized, helping to save room in the switching cabinet.

Safety is always first priority

Safety for people and for the system is the most important factor with all our developments. Here, this prerequisite has been met with a comfortable connection on the rear side. It allows for a safe connection from the circuit breakers to the busbar adapters. In addition, mounting times are significantly reduced. Modular system covers guarantee optimized shock protection all around, and thus the highest possible level of safety.



A system designed for worldwide use

Together with its system components, the SASY 60i busbar system is designed for worldwide use in control cabinets for mechanical and system engineering. Its design has even taken into account the greater clearance and creepage distances that must be observed in the U.S. pursuant to UL 508A.

For busbar applications that have not been type-tested, UL508A allows an ampacity of 1000A/inch² (1.55A/mm²). This value may be higher if the product or the application has been tested accordingly. Eaton has conducted extensive tests for the user's maximum benefit in using SASY 60i busbar systems. The advantage of such tests is that one can use busbar systems designed for higher rated currents than the default value allows. SASY 60i components and combinations are listed under File No. E300273 and E140305UL.



Since SASY 60i requires fewer system components, the need to stock parts and to place orders is diminished with the new Eaton busbar system.



SASY 60i Busbar System

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Technical Data

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SASY 60i Busbar System

- Components now also conforming to UL-standards for control systems
- 60 mm spacing between busbars
- 630, 1250 and 1600 A rated current
- Adapter technology for NZM1 to 3
- Adapter technology for xStart
- Slide fuse equipment
- Connection technique

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xBoard

Systems up to 630 A for Flat Busbars

Poles Number	Max. Rated Oper- ational Current I _e (A)	Special Features	Utilisation	Designation	Notes Article No.	Units per Package
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Busbar Support

- Thermoplastic, silicone-free, chlorine-free
- Halogen-free
- Self-extinguishing according to UL 94
- RAL 7035
- Track resistance CTI 200
- Temperature-resistant up to 120°C

IEC Busbar Support

3	630	With snap-in slide for adapting to the respective size of the bar	12 x 5/10 15 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10	BBS-3/FL 107066	With pre-drilled holes inside for screw-fixing	10 units
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4	630	With snap-in slide for adapting to the respective size of the bar	12 x 5/10 15 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10	BBS-4/FL 138381	With pre-drilled holes inside for screw-fixing	10 units
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UL Busbar Support

3	630	With snap-in slide for adapting to the respective size of the bar	12 x 5/10 20 x 5/10 30 x 5/10	BBS-3/FL-NA 107067	With pre-drilled holes inside for screw-fixing	10 units
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If used in feeder circuits according to UL 508A up to 600 V, it is necessary use the BBC-BT-NA base plate in addition.

PE/N Earth/Neutral Busbar Support

2	630	With snap-in slide for adapting to the respective size of the bar	12 x 5/10 15 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10	BBS-2/FL 107069	Can be mounted individually	10 units
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1	630	With snap-in slide for adapting to the respective size of the bar	12 x 5/10 20 x 5/10 30 x 5/10	BBS-1/FL 107161	Can be mounted individually	10 units
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Compact Busbar Support

3	360	With a removable contact block to adjust it to the respective size of the bar	12 x 5/10	BBS-3/FL-C 138370	With pre-drilled holes inside for screw-fixing and integrated end covers	10 units
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VT35310, VT35410



	Poles	Max. Rated Oper- ational Current I _e (A)	Special Features	Utilisation	Designation	Notes	Units per Package
	Number				Article No.		
	End Cover						
	–	–	–	To cover the busbar ends for BBS-3/FL and BBS-3/FL-NA	ES-BBS-3/FL 107068		10 units
	UL Base Plate						
	• Silicone-free, chlorine-free • Self-extinguishing according to UL 94 • Temperature-resistant up to 110°C						
	–	–	To be used when the air gap between fully equipped busbar systems and mounting plate is insufficient	Necessary for UL support BBS-3/FL-NA	BBC-BT-NA 107172	1100 mm long	2 units
	Busbar Covers						
	• Silicone-free, chlorine-free • Self-extinguishing according to UL 94 • Temperature-resistant up to 110°C						
	–	–	–	12 x 5 15 x 5 20 x 5 25 x 5 30 x 5	BBC-FL5 107173	12-30x5 1000 mm long	10 units
	–	–	–	12 x 10 15 x 10 20 x 10 25 x 10 30 x 10	BBC-FL10 107174	12-30x10 1000 mm long	10 units
	Max. Rated Operational Current I _e (A)		Dimension (mm x mm)	Length (mm)	Designation	Notes	Units per Package
					Article No.		
	Flat Copper Rails						
	160		12 x 5	1500	CU12X5 034121	tinned	10 units
				2250	CU12X5-2250 005093	tinned	10 units
	250		20 x 5	1500	CU20X5 044092	tinned	10 units
				2250	CU20X5-2250 007466	tinned	10 units
	460		20 x 10	1500	CU20X10 041719	tinned	10 units
				2250	CU20X10-2250 009839	tinned	10 units
	630		30 x 10	1500	CU30X10 051211	untreatedly	10 units
							

Systems up to 1250, 1600 A for Profile Bars						
Poles Number	Max. Rated Operational Current Ie (A)	Special Features	Utilisation	Designation Article No.	Notes	Units per Package
Busbar Support • Thermoplastic, silicone-free, chlorine-free • Halogen-free • Self-extinguishing according to UL 94 • RAL 7035 • Track resistance CTI 200 • Temperature-resistant up to 120°C						
Busbar Support Double-T-Profile						
	3	1600	Suitable as lateral and central support	Double-T-Profile	BBS-3/PR 107162	With pre-drilled 3 units holes inside for screw-fixing
	1	1600	Suitable for setting up a PE or N bar	Double-T-Profile	BBS-1/PR 107165	With pre-drilled 10 units holes inside for scew-fixing
End Cover						
	–	–	–	For the BBS-3/PR support	ES-BBS-3/PR 107164	4 units
UL Base Plate • Silicone-free, chlorine-free • Self-extinguishing according to UL 94 • Temperature-resistant up to 110°C						
	–	–	To be used when the air gap between fully equipped busbar systems and mounting plate is insufficient	Necessary for UL support BBS-3/FL-NA	BBC-BT-NA 107172	1100 mm long 2 units
Double-T-Profile Busbar						
	–	1250 ¹⁾	Tin-plated Cross-section 500 mm ²	For BBS-3/PR and BBS-1/PR supports	CU-BAR-500/T 107166	2400 mm long 1 unit
	–	1600 ¹⁾	Tin-plated Cross-section 720 mm ²	For BBS-3/PR and BBS-1/PR supports	CU-BAR-720/T 107167	2400 mm long 1 unit

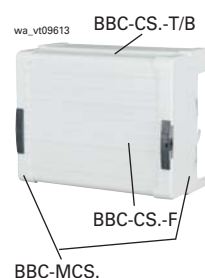
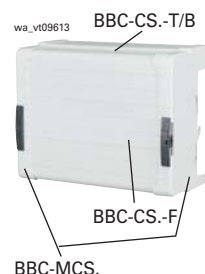
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Poles Number	Max. Rated Oper- ational Current I _e (A)	Special Features	Utilisation	Designation Article No.	Notes	Units per Package
Busbar Cover • Silicone-free, chlorine-free • Self-extinguishing according to UL 94 • Temperature-resistant up to 110°C						
–	–	–	For Double-T- profile	BBC-CU-BAR/PR 107175	1000 mm long	5 units

1) At a busbar temperature of 87.5°C and an ambient temperature of 35°C, please refer to the current load diagram in the Technical Data section for further values.

Covers for 630, 1250 and 1600 A Systems				
	Utilisation	Designation Article No.	Notes	Units per Package
	Spare Section Cover - Modular			
	To cover the front of the 60 mm system	BBC-RCOV1 107178	1100 mm long. To be used with BBC-MRCOV1 support only	2 units
	Support for Spare Section Cover			
	Suits any thickness of bars	BBC-MRCOV1 107179	To be used with spare section cover BBC-RCOV1 only	10 units
	Cover complete			
	For 3-pole systems	BBC-CS1 107209	228 mm long	1 unit
	For 3-pole systems	BBC-CS3 138377	270 mm long	1 unit
	For 4-pole systems	BBC-CS4 138387	228 mm long	1 unit
Single covers				
	Compartment Section Double-T			
	For 3-pole systems with BBS-3/PR	BBC-CS48/PR 107176	48 mm high 2400 mm long To be fixed at the (profile) bar support	1 unit
	For 3-pole systems with BBS-3/PR	BBC-CS76/PR 107177	76 mm high 2400 mm long To be fixed at the (profile) bar support	1 unit
	Front Plate Cover for front plate cut-out			
	Cover module for cut-out Height = 195 mm	AM-195/54 107963	54 mm width	15 units



Utilisation	Designation Article No.	Notes	Units per Package
System Cover - Kit - silicone-free, chlorine-free - Self-extinguishing according to UL 94 - Temperature-resistant up to 120°C			
Cover Profile - Front			
For 3-pole systems	BBC-CS2-F 107180	1100 mm long	1 unit
Cover Profile - Top/Bottom			
For 3-pole systems	BBC-CS2-T/B 107181	1100 mm long	2 units
Support Set for Cover Profile			
For 3-pole systems	BBC-MCS2 107182	1 set includes a right and left side support	1 unit
Cover Profile - Front			
For 4-pole systems	BBC-CS4-F 138834	1100 mm long	1 unit
Cover Profile - Top/Bottom			
For 4-pole systems	BBC-CS4-T/B 138383	1100 mm long	2 units
Support Set for Cover Profile			
For 4-pole systems	BBC-MCS4 138382	1 set includes a right and left side support	1 unit
Utilisation	Designation Article No.	Notes	Units per Package
System Cover - Compact			
Empty-section Cover, Modular - To cover the front of the Compact System - For use with BBC-MRCOV3-C only			
	BBC-RCOV3-C 138371	1100 mm long	2 units
Support for Empty-section Cover - Suitable for 5 and 10 mm bar thickness - For use with BBC-RCOV3-C only			
	BBC-MRCOV3-C 138372	12 x 5/10	10 units

Feeder Circuit Adapters for 630, 1250 and 1600 A Systems

Poles Number	Max. Rated Operational Current I _e (A)	Type of Conductor ¹⁾	Utilisation	Designation Article No.	Notes	Units per Package
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Connecting Terminal Plates

3	80	1.5 - 16 mm ² AWG 16 - AWG 6. 	12x5/10 15x5/10 20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP3/16 107183	20 mm wide. With spring-type terminal technology.	1 unit
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3	300	6 - 50 mm ² AWG 10 - AWG 2/0. 6x9x0.8	12x5/10 15x5/10 20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP3/50 107183	54 mm wide. Terminals can be removed for connecting non-cut conductors. Looping them through is possible. Termination space 10 x 15 mm.	1 unit
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3	440	35 - 120 mm ² AWG 2 - MCM 250. 10x16x0.8	12x5/10 15x5/10 20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP3/120 107184	81 mm wide. Terminals can be removed for connecting non-cut conductors. Looping them through is possible. Termination space 15 x 15 mm.	1 unit
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Connecting Terminal Plates Compact

3	480	35 - 120 mm ² AWG 2 - MCM 300. 10x20x1	12x5/10 15x5/10 20x5/10	BBA-TP3/100-C 138373	90 mm wide. Terminals can be removed for connecting non-cut conductors. Contacting is provided for through the cable bed. Compact System.	1 unit
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- ¹⁾
- Round conductor, single-wired
 - Round conductor, fine-wired with expertly pressed wire end ferrule
 - Round conductor, multi-wired
 - Sector conductor, single-wired
 - Sector conductor, multi-wired
 - Cu-Band
 - Cu-Bar

Feeder Circuit Adapters for 630, 1250 and 1600 A Systems

Poles	Max. Rated Operational Current I _e (A)	Type of Conductor ¹⁾	Utilisation	Designation Article No.	Notes	Units per Package
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Connecting Set with Cover 3-pole

- Silicone-free, chlorine-free
- Self-extinguishing according to UL 94
- Track resistance CTI 200
- Temperature-resistant up to 120°C

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3	560	120 - 300 mm ² MCM300 - MCM600. 	20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP3/300 107185	180 - 240 mm wide. Clearance between poles can be adjusted as required. To be fixed directly on top of the busbar terminal. Incl. cover cap in flexible width. Looping through is possible.	1 unit
3	800	Up to 10x32x1 30x25	20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP3/CU-BAND 107186	180 - 240 mm wide. Clearance between poles can be adjusted as required. To be fixed directly on top of the busbar terminal. Incl. cover cap in flexible width. Looping through is possible. Termination space 32 x 25 mm.	1 unit
3	1600	Up to (2x)10x50x1 Up to (2x)50x10	30x10 Double-T-Profile	BBA-TP3/1000 107207	228 mm wide. Co-ordinated up for Eaton NZM4. To be fixed directly on top of the busbar terminal. Incl. cover cap. Looping through is possible. Termination space 5 x 28 mm.	1 unit

- ¹⁾
- Round conductor, single-wired
 - Round conductor, fine-wired with expertly pressed wire end ferrule
 - Round conductor, multi-wired
 - Sector conductor, single-wired
 - Sector conductor, multi-wired
 - Cu-Band
 - Cu-Bar

Feeder Circuit Adapters for 630, 1250 and 1600 A Systems

Poles	Max. Rated Operational Current I _e (A)	Type of Conductor ¹⁾	Utilisation	Designation Article No.	Notes	Units per Package
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Connecting Set with Cover 4-pole

- Silicone-free, chlorine-free
- Self-extinguishing according to UL 94
- Track resistance CTI 200
- Temperature-resistant up to 120°C

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4	560	120 - 300 mm ² MCM300 - MCM600. 	20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP4/300 138385	180 - 228 mm wide. Clearance between poles can be adjusted as required. To be fixed directly on top of the busbar terminal. Incl. cover cap in flexible width. Looping through is possible.	1 unit
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4	800	Up to 10x32x1 30x25	20x5/10 25x5/10 30x5/10 Double-T-Profile	BBA-TP4/CU-BAND 138386	180 - 228 mm wide. Clearance between poles can be adjusted as required. To be fixed directly on top of the busbar terminal. Incl. cover cap in flexible width. Looping through is possible. Termination space 32 x 25 mm.	1 unit
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- ¹⁾
- Round conductor, single-wired
 - Round conductor, fine-wired with expertly pressed wire end ferrule
 - Round conductor, multi-wired
 - Sector conductor, single-wired
 - Sector conductor, multi-wired
 - Cu-Band
 - Cu-Bar

Terminals for 630, 1250 and 1600 A Systems

Max. Rated Oper- ational Current I _e (A)	Type of ¹⁾ Conductor	Special Features	Utilisation	Designation Article No.	Notes	Units per Package
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Brace Terminals

- Connection method to busbars without drilling

480	35 - 150mm ² , AWG2/0 - MCM300. ⊙ directly terminated, ⊙ ⊙	Connection method to busbars without drilling	12x5/10 20x5/10	AKS150 138374	Contacting of wire and busbar via a cable bed	6 units
500	95 - 185mm ² , AWG3/0 - MCM350. ⊙ directly terminated, ⊙ ⊙	Connection method to busbars without drilling	20x5/10 25x5/10 30x5/10 Double-T-Profile	AKS185 107195	Contacting of wire and busbar via a cable bed	6 units
600	150 - 300mm ² , MCM300 - MCM600. ⊙ directly terminated, ⊙ ⊙	Connection method to busbars without drilling	20x5/10 25x5/10 30x5/10 Double-T-Profile	AKS300 107196	Contacting of wire and busbar via a cable bed	3 units
800	⊞ 3x20x1 to 2x(10x32x1) ■ 32x25	Connection method to busbars without drilling. Termination space 32 x 25 mm.	20x5/10 25x5/10 30x5/10 Double-T-Profile	AKS-CU-BAND 107197	Contacting of wire and busbar via a contacting block	3 units
1600	Up to ⊞(2x)10x50x1 Up to ■(2x)50x10	Connection method to busbars without drilling. Termination space 55 x 28 mm.	20x5/10 25x5/10 30x5/10 Double-T-Profile	AKS1000 107208	Contacting of wire and busbar via a contacting block	1 unit
1600	Up to ■(2x)60x10	Connection method to busbars without drilling. Termination space 68 x 28 mm.	30x10 Double-T-Profile Triple-T-Profile	AKS1200 138375	Contacting of wire and busbar via a contacting block	3 units
1600	Up to ■(2x)100x10	Connection method to busbars without drilling. Termination space 105 x 28 mm.	30x10 Double-T-Profile Triple-T-Profile	AKS2000 138376	Contacting of wire and busbar via a contacting block	3 units

- ¹⁾ ⊙ Round conductor, single-wired
 ⊙ Round conductor, fine-wired with expertly pressed wire end ferrule
 ⊙ Round conductor, multi-wired
 ⊙ Sector conductor, single-wired
 ⊙ Sector conductor, multi-wired
 ⊞ Cu-Band
 ■ Cu-Bar

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VT13406



VT13206



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Terminals for 630, 1250 and 1600 A Systems

Max. Rated Oper- ational Current I _e (A)	Type of ¹⁾ Conductor	Special Features	Utilisation	Designation Article No.	Notes	Units per Package
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Profile Terminals ²⁾

- Connection method to busbars without drilling
- In case of parallel connection of two multi-layer copper bars, please place spacers in between

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1600	750 mm ² , Termination space 51 x 5-28 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP750 138364	Width 82 mm	3 units
1600	800 mm ² , Termination space 41 x 20-42 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP800 107198	Width 72 mm	3 units
1600	900 mm ² , Termination space 64 x 5-28 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP900 138365	Width 94 mm	3 units
1600	1000 mm ² , Termination space 51 x 20-42 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP1000 107199	Width 94 mm	3 units
2000	1200 mm ² , Termination space 64 x 20-42 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP1200 138366	Width 94 mm	3 units
2500	1600 mm ² , Termination space 81 x 20-42 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP1600 138367	Width 112 mm	3 units
3000	2000 mm ² , Termination space 101 x 20-42 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP2000 138368	Width 132 mm	3 units
3200	3600 mm ² , Termination space 101 x 23-45 ■ ■ ■ ■ ■	Connection method to busbars without drilling	Double-T- Profile	AKP3600 138369	Width 132 mm	3 units

- ¹⁾ ○ Round conductor, single-wired
 ● Round conductor, fine-wired with expertly pressed wire end ferrule
 ⊙ Round conductor, multi-wired
 ◊ Sector conductor, single-wired
 ⊠ Sector conductor, multi-wired
 ■ Cu-Band
 ■ Cu-Bar

²⁾ For a UL508A system with profile terminals you need to use the UL base plate BBC-BT-NA and the busbar cover BBC-CU-BAR/PR.

Terminals for 630, 1250 and 1600 A Systems



Max. Rated Oper- ational Current I _e (A)	Type of Conductor ¹⁾	Special Features	Utilisation	Designation Article No.	Units per Package
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Universal Conductor Terminal 5 mm

180	1.5 - 16mm ² , AWG 14 - AWG 6. ⊗ directly terminated, ○ ⊙ ≡ 8x6x0.5	With integrated retaining spring, captive terminal screw, opened termination space 7.5 x 7.5 mm	All flat busbars of a thickness of 5 mm	AKU16/5 107187	100 units
270	4 - 35mm ² , AWG 10 - AWG 2. ⊗ directly terminated, ○ ⊙ ≡ 3x9x0.8 or 6x9x0.8	With integrated retaining spring, captive terminal screw, opened termination space 10.5 x 11 mm	All flat busbars of a thickness of 5 mm	AKU35/5 107188	50 units
400	16 - 70mm ² , AWG 4 - AWG 2/0. ⊗ directly terminated, ⊙ ≡ 2x(3x9x0.8) or 6x9x0.8	With integrated retaining spring, captive terminal screw, opened termination space 14 x 14 mm	All flat busbars of a thickness of 5 mm	AKU70/5 107189	25 units
440	16 - 120mm ² , AWG 4 - MCM 250. ⊗ directly terminated, ⊙ ≡ 4x16x0.8 or 6x16x0.8 or 10x16x0.8	With integrated retaining spring, captive terminal screw, opened termination space 17 x 15 mm	All flat busbars of a thickness of 5 mm	AKU120/5 107190	25 units

- ¹⁾ ○ Round conductor, single-wired
 ⊗ Round conductor, fine-wired with expertly pressed wire end ferrule
 ⊙ Round conductor, multi-wired
 ◇ Sector conductor, single-wired
 ⊠ Sector conductor, multi-wired
 ≡ Cu-Band
 ■ Cu-Bar

Terminals for 630, 1250 and 1600 A Systems



Max. Rated Oper- ational Current I _e (A)	Type of Conductor ¹⁾	Special Features	Utilisation	Designation Article No.	Units per Package
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Universal Connection Terminal 10 mm

180	1.5 - 16mm ² , AWG 14 - AWG 6. ⊗ directly terminated, ○ ⊙ ≡ 8x6x0.5	With integrated retaining spring, captive terminal screw, opened termination space 7.5 x 7.5 mm	All flat busbars of a thickness of 10 mm	AKU16/10 107191	100 units
270	4 - 35mm ² , AWG 10 - AWG 2. ⊗ directly terminated, ○ ⊙ ≡ 3x9x0.8 or 6x9x0.8	With integrated retaining spring, captive terminal screw, opened termination space 10.5 x 11 mm	All flat busbars of a thickness of 10 mm	AKU35/10 107192	50 units
400	16 - 70mm ² , AWG 4 - AWG 2/0. ⊗ directly terminated, ⊙ ≡ 2x(3x9x0.8) or 6x9x0.8	With integrated retaining spring, captive terminal screw, opened termination space 14 x 14 mm	All flat busbars of a thickness of 10 mm	AKU70/10 107193	25 units
440	16 - 120mm ² , AWG 4 - MCM 250. ⊗ directly terminated, ⊙ ≡ 4x16x0.8 or 6x16x0.8 or 10x16x0.8	With integrated retaining spring, captive terminal screw, opened termination space 17 x 15 mm	All flat busbars of a thickness of 10 mm	AKU120/10 107194	25 units
490	Cable lug M8	With integrated retaining spring, captive terminal screw, opened termination space, bolt M8x8	All flat busbars of a thickness of 10 mm, Double-T-profile	AKU-M8/10 138362	20 units
630	Cable lug M10	With integrated retaining spring, captive terminal screw, opened termination space, bolt M10x10	All flat busbars of a thickness of 10 mm, Double-T-profile	AKU-M10/10 138361	6 units

- ¹⁾ ○ Round conductor, single-wired
 ⊗ Round conductor, fine-wired with expertly pressed wire end ferrule
 ⊙ Round conductor, multi-wired
 ◊ Sector conductor, single-wired
 ◊ Sector conductor, multi-wired
 ≡ Cu-Band
 ■ Cu-Bar

Terminals for 630, 1250 and 1600 A Systems

Max. Rated Oper- ational Current I _e (A)	Type of Conductor ¹⁾	Special Features	Utilisation	Designation Article No.	Units per Package
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Plattenklemmen

630	–	Width 50 mm	All flat busbars of a thickness of 10 mm	PK900 138378	3 units
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Anschlussklemmen

630	95 - 300mm ²	Width 48 mm. Kontaktierung der Leitung mit der Sammel- schiene erfolgt über Kabelbett.	30 x 10mm Double-T-profile Triple-T-profile	AK300 138336	3 units
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¹⁾  Round conductor, single-wired
 Round conductor, fine-wired with expertly pressed wire end ferrule
 Round conductor, multi-wired
 Sector conductor, single-wired
 Sector conductor, multi-wired
 Cu-Band
 Cu-Bar

Lengthwise Bar Connections for 630, 1250 and 1600 A Systems

Max. Rated Oper- ational Current I _e (A)	Width (mm)	Special Features	Utilisation	Designation Article No.	Notes	Units per Package
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Busbar Connecting Terminals

- For drill-free connection of identical types of busbars

wa_vt27113



630	38	For identically shaped, flat copper bars	12 x 5/10 15 x 5/10 20 x 5/10	BBT-CU12-20X5/10-38 138379	Spacing between systems 100 - 110 mm. Max. permissible mis-alignment of bars is 1 mm	12 units
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wa_vt12513



630	150	For identically shaped, flat copper bars	12 x 5/10 15 x 5/10 20 x 5/10	BBT-CU12-20X5/10-150 107200	Spacing between systems 100 - 110 mm. Max. permissible mis-alignment of bars is 1 mm	3 units
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01063549_0



630	40	For identically shaped, flat copper bars	12 x 5/10 15 x 5/10 20 x 5/10	BBT-CU20-30X5/10-40 138380	Spacing between systems 50 - 60 mm. Max. permissible mis-alignment of bars is 5 mm	6 units
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wa_vt12113



630	95	For identically shaped, flat copper bars	12 x 5/10 15 x 5/10 20 x 5/10	BBT-CU20-30X5/10-95 107201	Spacing between systems 50 - 60 mm. Max. permissible mis-alignment of bars is 5 mm	3 units
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wa_vt12013



630	150	For identically shaped, flat copper bars	12 x 5/10 15 x 5/10 20 x 5/10	BBT-CU20-30X5/10-150 107202	Spacing between systems 100 - 110 mm. Max. permissible mis-alignment of bars is 5 mm	3 units
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wa_vt11913



1600	50	For different and identical types of double-T-profile bars	Double-T-profile	BBT-CU-BAR500/720-50 107203	Spacing between systems 9 - 20 mm. Max. permissible mis-alignment of bars is 2 mm	6 units
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wa_vt11813



1600	150	For different and identical types of double-T-profile bars	Double-T-profile	BBT-CU-BAR500/720-150 107204	Spacing between systems 100 - 110 mm. Max. permissible mis-alignment of bars is 5 mm	3 units
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NZM Busbar Adapter, 3-pole ¹⁾

Max. Rated Oper. Current I _e (A)	Rated Operational Voltage U _e (V)	Adapter Width (mm)	Adapter Length (mm)	Special Features	Utili- sation	Designation Article No.	Notes	Units per Package
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Busbar Adapter NZM

- For use on flat copper bars 12 - 30 x 5/10, double-T-profiles and triple-T-profiles
- Self-extinguishing according to UL 94
- Track resistance CTI 200
- Temperature-resistant up to 120°C

1230PIC-668 AO



160	690	92	200	For connecting to the system at the top or bottom through fixed connection bars included in the scope of delivery ²⁾	NZM1 PN1 N1 NS1	NZM1-XAD160 104554	For switches with standard connection frame-type terminals. To be snapped onto the busbar by means of a combi-base.	1 unit
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wa_vt12213



250	690	106	190	For connecting to the system at the top/bottom through a tube-type of connection at the rear. Tube included in the scope of delivery.	NZM2 PN2 N2 NS2	NZM2-XAD250 104555	Use only in combination with auxiliary type (+)NZM2-XKR4. To be screwed onto the busbar by means of a claw-type of clamp.	1 unit
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wa_vt22513, wa_vt12213



630	690	140	300	For connecting to the system at the top/bottom through a tube-type of connection at the rear. Tube included in the scope of delivery.	NZM3 PN3 N3	NZM3-XAD630 107206	Use only in combination with auxiliary type (+)NZM3-XKR13. To be screwed onto the busbar by means of a claw-type of clamp.	1 unit
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Terminal for device adapter NZM

wa_vt12713



250	690	–	–	To cover the connection to the system at the top/bottom	NZM2 PN2 N2 NS2	NZM2-XKR4 281666	For device combination NZM2 use with auxiliary type +NZM2-XKR40 or +NZM2-XKR4U	1 unit
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wa_vt12613



630	690	–	–	To cover the connection to the system at the top/bottom	NZM3 PN3 N3	NZM3-XKR13 281668	For device combination NZM3 use with auxiliary type +NZM3-XKR130 or +NZM3-XKR13U	1 unit
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¹⁾ To be snapped onto the voltage-free busbar.

²⁾ Thanks to the combi-base it can be adjusted to a bar width of both 5 and 10 mm.

NZM Busbar Adapter, 4-pole ¹⁾

Max. Rated Oper. Current I _e (A)	Rated Operational Voltage U _e (V)	Adapter Width (mm)	Adapter Length (mm)	Special Features	Utili- sation	Designation Article No.	Notes	Units per Package
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Busbar Adapter NZM

- For use on flat copper bars 12 - 30 x 5/10, double-T-profiles and triple -T-profiles
- Self-extinguishing according to UL 94
- Track resistance CTI 200
- Temperature-resistant up to 120°C

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250	690	140	—	For connecting to the system at the top through a tube-type of connection at the rear. Tube included in the scope of delivery.	NZM2(-4) PN2(-4) 138388 N2(-4) NS2(-4)	NZM2-4-XAD250	Use only in combination with auxiliary type (+)NZM2-4-XKR4. To be screwed onto the busbar by means of a claw-type of clamp.	1 unit
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01063598_0



630	690	185	—	For connecting to the system at the top through a tube-type of connection at the rear. Tube included in the scope of delivery.	NZM3(-4) PN3(-4) 138389 N3(-4) NS3(-4)	NZM3-4-XAD630	Use only in combination with auxiliary type (+)NZM3-4-XKR13. To be screwed onto the busbar by means of a claw-type of clamp.	1 unit
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Terminal for Device Adapter NZM

NZM2-4-XKR4



250	690	—	—	To cover the connection to the system at the top	NZM2-4 PN2-4 118907 N2-4 NS2-4	NZM2-4-XKR4	For device combination NZM2 use with auxiliary type +NZM2-4-XKR40	1 unit
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NZM2-4-XKR4



630	690	—	—	To cover the connection to the system at the top	NZM3-4 PN3-4 119020 N3-4 NS3-4	NZM3-4-XKR13	For device combination NZM3 use with auxiliary type +NZM3-4-XKR130	1 unit
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¹⁾ To be snapped onto the voltage-free busbar.

²⁾ Thanks to the combi-base it can be adjusted to a bar width of both 5 and 10 mm, cross-section of conductor 6 x 9 x 0.8.

xStart Busbar Adaptor, 3-pole ¹⁾

Max. Rated Oper- ational Current I _e (A)	Rated Operational Voltage U _e (V)	Wire Cross Section	Adapter Width (mm)	Adapter Length (mm)	Support Rails	Utilisation	Designation Article No.	Notes	Units per Package
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xStart Busbar Adaptor 25 A

25	690	AWG12	45	200	1	PKZM0+ Contactor DIL M 7 Contactor DIL M 9 Contactor DIL M 12 Contactor DIL M 15 MSC-D-0.25-M7... to MSC-D-16-M15...	BBA0-25 101451	Direct starter set PKZM0-XDM12	4 units
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25	690	AWG12	90	200	1	PKZM0+ 2x Contactor DIL M 7-01 2x Contactor DIL M 9-01 2x Contactor DIL M 12-01 MSC-R-0.25-M7... to MSC-R-12-M12...	BBA0R-25 101453	Reversing starter set PKZM0-XRM12	2 units
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xStart Busbar Adaptor 25 A, Universal Type

25	690	AWG12	45	200	2	Support rail adjustable on the 1.25 mm grid	BBA0-25/2TS 101481		4 units
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xStart Busbar Adaptor 32 A

32	690	AWG10	45	200	2	PKZM0+ Contactor DIL M 17 Contactor DIL M 25 Contactor DIL M 32 MSC-D-16-M17... to MSC-D-32-M32...	BBA0-32 101452	Electrical contact module PKZM0-XM32 DE	4 units
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32	690	AWG10	90	200	3	PKZM0+ 2x Contactor DIL M 17-01 2x Contactor DIL M 25-01 2x Contactor DIL M 32-01 MSC-R-16-M17... to MSC-R-32-M32...	BBA0R-32 101454	Electrical contact module PKZM0-XM32 DE Reverse wiring set DILM32-XRL	2 units
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xStart Busbar Adaptor 32 A, for Spring-type Terminal

32	690	–	45	200	2	Support rail adjustable on the 1.25 mm grid	BBA0-32/2TS-C 116708	With spring-type terminal technology, up to 1.5-6 mm ² . For example for 1-phase applikations.	4 units
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¹⁾ Can be used with all busbars in a 60 mm system. Thanks to the combi-base it is suitable for both 5 mm and 10 mm thickness of the bar as well as for double-T-profile bars.
To be snapped onto the voltage-free busbar.

xStart Busbar Adaptor, 3-pole ¹⁾

Max. Rated Oper- ational Current I _e (A)	Rated Operational Voltage U _e (V)	Wire Cross Section	Adapter Width (mm)	Adapter Length (mm)	Support Rails	Utilisation	Designation Article No.	Notes	Units per Package
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xStart Busbar Adaptor 63 A

63	690	AWG8	72	260	2	PKZ2+ Contactor DIL M 7 Contactor DIL M 9 Contactor DIL M 12 Contactor DIL M 17 Contactor DIL M 25 Contactor DIL M 32 Contactor DIL M 40	BBA2L-63 101480		2 units
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63	690	AWG8	72	200	1	PKZ2	BBA2-63 101458		4 units
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63	690	AWG8	72	200	2	PKZ2+ Contactor DILM7 - DILM15 DILM17 - DILM38 DILM40 - DILM 65 Switch drive SE1A-PKZ2 and S-PKZ2 in connection with clipsplate C-PKZ2	BBA2-63/2TS 116900		4 units
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63	690	AWG8	55	260	2	PKZM4+ Contactor DIL M 17 Contactor DIL M 25 Contactor DIL M 32 Contactor DIL M 40 Contactor DIL M 50 Contactor DIL M 65	BBA4L-63 101459	Electrical connector for PKZM4+DILM40...65: PKZM4-XM65 DE	4 units
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63	690	AWG8	55	200	1	PKZM4	BBA4-63 101457		4 units
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xStart Busbar Adaptor 80 A

80	690	–	72	200	2	universal	BBA2-80/2TS-S 116901	with screw- type terminal technology up to AWG6, for example for 1-phase applications (not UL/CSA compatible without an additional component)	4 units
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¹⁾ Can be used with all busbars in a 60 mm system. Thanks to the combi-base it is suitable for both 5 mm and 10 mm thickness of the bar as well as for double-T-profile bars.
To be snapped onto the voltage-free busbar.

xStart Busbar Adaptor, 3-pole ¹⁾

Max. Rated Oper- ational Current I _e (A)	Rated Operational Voltage U _e (V)	Wire Cross Section	Adapter Width (mm)	Adapter Length (mm)	Support Utilisation Rails	Designation Article No.	Notes	Units per Package
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xStart Busbar Adaptor, Universal Type

–	–	–	45	200	2	Support rail adjustable on the 1.25 mm grid	BBA0/2TS-L 101482	Without electrical contacting, auxiliary to BBA... for the setup of reversing starters, for example	4 units
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–	–	–	54	260	2	Support rail adjustable on the 1.25 mm grid	BBA4/2TS-L 101483	Without electrical contacting, auxiliary to BBA... for the setup of reversing starters, for example	4 units
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Side Module

–	–	–	9	200	–		BBA-XSM 101484	Can be placed on both sides of BBA, to increase the add-on width	10 units
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Accessories

Width (mm)	Notes	Type Designation	Article No.	Units per package
Support rail				
45	Used for	PKZM0-XMR	239364	10 units
54	BBA...	PKZM0-XMR54	113911	10 units
72	adapter	PKZM0-XMR72	113912	10 units
Connecting cable				
	Used for	BBA-XLT-6-130	116902	30 units
	BBA with screw-type or spring-type terminals	BBA-XLT-16-142	116903	30 units

Double adapter Z-SS-60-ADD/6

- Cross section 6 mm² - I_e = 35 A

Width (mm) / MU / Cross section	Type Designation	Article No.	Units per package
45 / 2,5 / 6mm ²	Z-SS-60-ADD/6-45	288790	1 / 10 units
54 / 3 / 6mm ²	Z-SS-60-ADD/6-54	288791	1 / 10 units
72 / 4 / 6mm ²	Z-SS-60-ADD/6-72	288792	1 / 10 units
81 / 4,5 / 6mm ²	Z-SS-60-ADD/6-81	288793	1 / 10 units

¹⁾ Can be used with all busbars in a 60 mm system. Thanks to the combi-base it is suitable for both 5 mm and 10 mm thickness of the bar as well as for double-T-profile bars.
To be snapped onto the voltage-free busbar.

wa_vt11013



wa_vt111413



wa_vt111513



wa_vt130813



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wa_sg09804



Motorstarter
Betätigungsspannung
230 V 50 Hz

Motorstarter
Betätigungsspannung
24 V DC

Rated Oper- ational Power AC3 380 V 400 V 415 V P (kW)	Rated Oper- ational Current 400 V I _e (A)
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Units
per
PackageUnits
per
Package

0.06	0.21	PKZM0-0,25 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-0,25-M7(230V50Hz)/BBA 102737	1 unit	MSC-D-0,25-M7(24VDC)/BBA 102964	1 unit
0.09	0.31	PKZM0-0,4 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-0,4-M7(230V50Hz)/BBA 102738	1 unit	MSC-D-0,4-M7(24VDC)/BBA 102965	1 unit
0.12 0.18	0.41 0.6	PKZM0-0,63 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-0,63-M7(230V50Hz)/BBA 102739	1 unit	MSC-D-0,63-M7(24VDC)/BBA 102966	1 unit
0.25	0.8	PKZM0-1 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-1-M7(230V50Hz)/BBA 102950	1 unit	MSC-D-1-M7(24VDC)/BBA 102967	1 unit
0.37 0.55	1.1 1.5	PKZM0-1,6 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-1,6-M7(230V50Hz)/BBA 102951	1 unit	MSC-D-1,6-M7(24VDC)/BBA 102968	1 unit
0.75	1.9	PKZM0-2,5 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-2,5-M7(230V50Hz)/BBA 102952	1 unit	MSC-D-2,5-M7(24VDC)/BBA 102969	1 unit
1.1 1.5	2.6 3.6	PKZM0-4 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-4-M7(230V50Hz)/BBA 102953	1 unit	MSC-D-4-M7(24VDC)/BBA 102970	1 unit
2.2	5	PKZM0-6,3 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-6,3-M7(230V50Hz)/BBA 102954	1 unit	MSC-D-6,3-M7(24VDC)/BBA 102971	1 unit
3	6.6	PKZM0-10 +DILM7-10 +PKZM0-XDM12 +BBA0-25	MSC-D-10-M7(230V50Hz)/BBA 102955	1 unit	MSC-D-10-M7(24VDC)/BBA 102972	1 unit
4	8.5	PKZM0-10 +DILM9-10 +PKZM0-XDM12 +BBA0-25	MSC-D-10-M9(230V50Hz)/BBA 102956	1 unit	MSC-D-10-M9(24VDC)/BBA 102973	1 unit
5.5	11.3	PKZM0-12 +DILM12-10 +PKZM0-XDM12 +BBA0-25	MSC-D-12-M12(230V50Hz)/BBA 102957	1 unit	MSC-D-12-M12(24VDC)/BBA 102974	1 unit
7.5	15.2	PKZM0-16 +DILM17-10 +PKZM0-XM32 +BBA0-32	MSC-D-16-M17(230V50Hz)/BBA 102961	1 unit	MSC-D-16-M17(24VDC)/BBA 102978	1 unit
11	21.7	PKZM0-25 +DILM25-10 +PKZM0-XM32 +BBA0-32	MSC-D-25-M25(230V50Hz)/BBA 102962	1 unit	MSC-D-25-M25(24VDC)/BBA 102979	1 unit
15	29.3	PKZM0-32 +DILM32-10 +PKZM0-XM32 +BBA0-32	MSC-D-32-M32(230V50Hz)/BBA 102963	1 unit	MSC-D-32-M32(24VDC)/BBA 102980	1 unit

^{*)} Technical details see Eaton main catalogue Industrial Switch Gear

1) Can be used with all busbars in a 60 mm system. Thanks to the combi-base it is suitable for both 5 mm and 10 mm thickness of the bar as well as for double-T-profile bars.
To be snapped onto the voltage-free busbar.



Symbol photo

xStart-Busbar Adaptor, 3-pole¹⁾, MSC-R.../BBA



Symbol photo

Motorstarter Betätigungsspannung 230 V 50 Hz				Motorstarter Betätigungsspannung 24 V DC		
Motor data						
Rated Oper- ational Power AC3 380 V 400 V 415 V P (kW)	Rated Oper- ational Current 400 V Ie (A)	Components	Designation Article No.	Units per Package	Designation Article No.	Units per Package
*)						
0.06	0.21	PKZM0-0,25 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-0,25-M7(230V50Hz)/BBA 102981	1 unit	MSC-R-0,25-M7(24VDC)/BBA 102997	1 unit
0.09	0.31	PKZM0-0,4 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-0,4-M7(230V50Hz)/BBA 102982	1 unit	MSC-R-0,4-M7(24VDC)/BBA 102998	1 unit
0.12 0.18	0.41 0.6	PKZM0-0,63 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-0,63-M7(230V50Hz)/BBA 102983	1 unit	MSC-R-0,63-M7(24VDC)/BBA 102999	1 unit
0.25	0.8	PKZM0-1 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-1-M7(230V50Hz)/BBA 102984	1 unit	MSC-R-1-M7(24VDC)/BBA 103000	1 unit
0.37 0.55	1.1 1.5	PKZM0-1,6 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-1,6-M7(230V50Hz)/BBA 102985	1 unit	MSC-R-1,6-M7(24VDC)/BBA 103001	1 unit
0.75	1.9	PKZM0-2,5 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-2,5-M7(230V50Hz)/BBA 102986	1 unit	MSC-R-2,5-M7(24VDC)/BBA 103002	1 unit
1.1 1.5	2.6 3.6	PKZM0-4 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-4-M7(230V50Hz)/BBA 102987	1 unit	MSC-R-4-M7(24VDC)/BBA 103003	1 unit
2.2	5	PKZM0-6,3 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-6,3-M7(230V50Hz)/BBA 102988	1 unit	MSC-R-6,3-M7(24VDC)/BBA 103004	1 unit
3	6.6	PKZM0-10 +2xDILM7-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-10-M7(230V50Hz)/BBA 102989	1 unit	MSC-R-10-M7(24VDC)/BBA 103005	1 unit
4	8.5	PKZM0-10 +2xDILM9-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-10-M9(230V50Hz)/BBA 102990	1 unit	MSC-R-10-M9(24VDC)/BBA 103006	1 unit
5.5	11.3	PKZM0-12 +2xDILM12-01 +PKZM0-XMR12 +BBA0R-25	MSC-R-12-M12(230V50Hz)/BBA 102991	1 unit	MSC-R-12-M12(24VDC)/BBA 103007	1 unit
7.5	15.2	PKZM0-16 +2xDILM17-01 +PKZM0-XMR32 +DILM32-XRL	MSC-R-16-M17(230V50Hz)/BBA 102994	1 unit	MSC-R-16-M17(24VDC)/BBA 103010	1 unit
11	21.7	PKZM0-25 +2xDILM25-01 +PKZM0-XMR32 +DILM32-XRL	MSC-R-25-M25(230V50Hz)/BBA 102995	1 unit	MSC-R-25-M25(24VDC)/BBA 103011	1 unit
15	29.3	PKZM0-32 +2xDILM32-01 +PKZM0-XMR32 +DILM32-XRL	MSC-R-32-M32(230V50Hz)/BBA 102996	1 unit	MSC-R-32-M32(24VDC)/BBA 103012	1 unit

*) Technical details see Eaton main catalogue Industrial Switch Gear

¹⁾ Can be used with all busbars in a 60 mm system. Thanks to the combi-base it is suitable for both 5 mm and 10 mm thickness of the bar as well as for double-T-profile bars.
To be snapped onto the voltage-free busbar.

Slide Fuse Equipment, 3-pole

Max. Rated Oper- ational Current I _e (A)	Rated Voltage U _e (V AC)	Component Size	Width	Utilisation	Designation Article No.	Notes	Units per Package
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D-Type Slide Fuse-Base

- Incl. shock hazard protection cover with front and bottom plate and marking label
- Delivered empty, without screw caps

63	400	E18, D02	27	12 x 5/10 20 x 5/10 25 x 5/10	D02-SO/63/3-R-27 114315	Cartridge-ring adapter insert	10 units
			36	30 x 5/10 Double-T	Z-D02/R/3-36 100663	Cartridge-ring adapter insert	60 units
			54		Z-D02/R/3-54 100664	Cartridge-ring adapter insert	40 units

25	500	E27, D II	45	12 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10 Double-T	DII-SO/25/3-R 107965	Gauge ring	10 units
					DII-SO/25/3-R-PS 110394	Screw-in gauge ring	10 units

63	690	E33, D III	54	12 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10 Double-T	DIII-SO/63/3-R 107966	Gauge ring	10 units
					DIII-SO/63/3-R-PS 110395	Screw-in gauge ring	10 units

D-Type Slide Fuse-Base, compact-type

- Incl. shock hazard protection cover with front and bottom plate and marking label
- Delivered empty, without screw caps

63	400	E18, D02	36	12 x 5/10	FCFBD02BBC60-3-36 139532	Cartridge-ring adapter insert, snap-on mechanism when sliding it onto the busbar.	6 units
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Covers

Set for covering busbar support			36	D02	Z-D02-S-AB-SET 100662	Suitable for D02-SO/63/3-R-27	10 units
Side cover			–	DII	SBS-RS60 060541	Suitable for DII.-SO/.../3-R(-PS)	10 units

SG82411



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SG62812



SG60412



wa_sg01713













Slide Fuse Equipment, 3-pole							
Max. Rated Oper- ational Current Ie (A)	Rated Voltage Ue (V AC)	Component Size	Width	Utilisation	Designation Article No.	Notes	Units per Package
Screw Caps							
63	400	E18, D02	–	D02-SO...	Z-D02/SK 100651	–	20/500
25	500	E27, D II	–	DII-SO...	Z-DII/SK 112148	–	50/600
63	500	E33, D III	–	DIII-SO...	Z-DIII/SK 112149	–	30/360
63	690	E33, D III	–	DIII-SO...	Z-DIII/SK-690 118904	–	3
Adapter Spring							
• to accommodate D01 fuse-links in Z-D02/SK screw caps							
16	–	D02-D01	–	–	Z-D02/SIKA-HF– 263149		50/3000
Accessories: For fuse-links, cartridge-ring adapter inserts, gauge rings and screw-in gauge rings see chapter Accessories Fuse Devices							
D02 Switch-Disconnect-Fuse							
• Incl. shock hazard protection cover with front and bottom plate							
• Delivered empty, without screw caps							
63	400	E18, D02	36	20 x 5/10 30 x 5/10 Double-T	D02-S/63/3-RS 284649	Cartridge-ring adapter insert	10 units
Screw Cap							
63	400	E18, D02	–	D02-SO...	Z-D02/SK 100651	–	20/500
Adapter Spring							
• to accommodate D01 fuse-links in Z-D02/SK screw caps							
16	–	D02-D01	–	–	Z-D02/SIKA-HF– 263149		50/3000
Accessories: For fuse-links, cartridge-ring adapter inserts see chapter Accessories Fuse Devices							

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wa_sg02612



SG45912



wa_sg04013



wa_sg02612



Slide Fuse Equipment, 3-pole

Rated Operational Current I _e (A)	Rated Voltage U _e (V AC)	Size	Width	Utilisation	Designation Article No.	Notes	Units per Package
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Switch-Disconnecter-Fuse D02 (+D01) + C

- Visual tripping indicator is flashing
- Delivered empty, without cartridge-ring adapter inserts and fuse-links
- Delivered with adapter springs for fuse-links D01 or cylindrical fuse-links 10x38
- Contact position indicator red - green
- Plug-in technique without screw caps
- All-pole and hand independent switching of load
- Version D02-LTS/63/3-R-HK with incorporated auxiliary switch
- Lead-seal- and lockable

3P

max. 63	400	E18, D02	27	12 x 5/10	D02-LTS/63/3-R	Cartridge-ring	3
max. 32	400	C 10x38		15 x 5/10	114316	adapter insert	
				20 x 5/10		without auxiliary	
				25 x 5/10		switch	
				30 x 5/10			
				Double-T			
					D02-LTS/63/3-R-HK	Cartridge-ring	3
					114318	adapter insert	
						with auxiliary	
						switch	

3P+N

max. 63	400	E18, D02	27	12 x 5/10	D02-LTS/63/3N-R	Cartridge-ring	3
max. 32	400	C 10x38		15 x 5/10	114317	adapter insert	
				20 x 5/10		without auxiliary	
				25 x 5/10		switch	
				30 x 5/10			
				Double-T			
					D02-LTS/63/3N-R-HK	Cartridge-ring	3
					114319	adapter insert	
						with auxiliary	
						switch	

Accessories for D02-LTS/63..

- D0** Fuse-links Z-D0./SE-...
Cartridge-ring adapter inserts D01: Z-D02-D01/PE-...
D02: Z-D02/PE-...
Adapter spring Z-D02-LTS-HF (scope of delivery)

- C** Fuse-links Z-C10/SE-...
Adapter spring Z-D02-LTS-HF (scope of delivery)

See chapter Accessories Fuse Devices

Adapter Spring

- to accommodate D01 fuse-links or cylindrical fuse-links 10x38 in the Switch-disconnector-fuse D02-LTS/63...

16	–	D02-D01	–	–	Z-D02-LTS-HF	–	12 / 288
32		C 10x38			114323		

Accessories: For fuse-links, cartridge-ring adapter inserts see chapter Accessories Fuse Devices

SG82311



SG82211



SG81811



Slide Fuse Equipment, 3-pole

Rated Operational Current I _e (A)	Max. Fuse-link 500V (A)	690V (A)	Size	Utilisation	Designation Article No.	Notes	Units per Package
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NH-Fuse-Switch-Disconnecter

- Incl. shock hazard protection at the top and bottom (except with GST00-160-40-60-AOU, GST00-160-40-60-AOU-F)
- Drill-free mounting

SG45812



100	100	–	000	20 x 5/10 30 x 5/10 Double-T	LTS-100/C00/3-R 284690	Width 63 mm. Connection at the bottom. Lift terminal 1.5-50mm ²	1
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SG45712



160	160	100	00	20 x 5/10 25 x 5/10 30 x 5/10 Double-T	GST00-160-40-60-AOU 224550	Connection at the top or bottom. Lift Terminal 1.5-50mm ² No shock hazard	1
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For ordering information on shock hazard protection BS-SET-GST00 see "Accessories" protection.

SG45612



160	160	100	00	20 x 5/10 25 x 5/10 30 x 5/10 Double-T	GST00-160-40-60-AOU-F 149418	Connection at the top or bottom. Screw M8.	1
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For ordering information on shock hazard protection BS-SET-GST00 see "Accessories" protection.

SG46512



250	250	200	1	20 x 5/10 25 x 5/10 30 x 5/10 Double-T	GST1-AO 107250	Connection at the top. Screw M10.	1
					GST1-AU 107251	Connection at the bottom. Screw M10.	1

SG46412



400	400	315	2	20 x 5/10 25 x 5/10 30 x 5/10 Double-T	GST2-AO 107252	Connection at the top. Screw M10.	1
					GST2-AU 107253	Connection at the bottom. Screw M10.	1

SG46212



630	630	500	3	20 x 5/10 25 x 5/10 30 x 5/10 Double-T	GST3-AO 107254	Connection at the top. Screw M10.	1
					GST3-AU 107255	Connection at the bottom. Screw M10.	1

Accessories for NH-Fuse-Switch-Disconnecter

Shock Hazard Protection Set GST00

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–	–	–	–	GST00...	BS-SET-GST00 107955	1 set includes shock hazard protection at the top and bottom.	1
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For NH-fuse-links Z-NH... and solid-links Z-NH-.../TR see chapter Accessories Fuse Devices.



Slide Fuse Equipment, 3-pole

Max. Rated Oper- ational Current Ie (A)	Max. Fuse-link 500V (A)	690V (A)	Component Size	Utilisation	Designation Article No.	Notes	Units per Package
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NH-Fuse-Switch-Disconnecter

- Incl. shock hazard protection at the top and bottom
- Drill-free mounting
- For mounting onto busbars with distance between busbars 60 mm

100	100	100	NH00(0) (width max. 21 mm)	12 x 5/10	FCFSDNH00BBC60-3 139533	-	1
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Slide Fuse Equipment, 3-pole						
	Rated Oper- ational Voltage	Component Size	Utilisation	Designation Article No.	Notes	Units per Package
	Switch cover with Fuse Monitoring for NH-Fuse-Switch-Disconnecter					
	- Operating indication 1 green LED Error indication 3 red LEDs (F1, F2, F3) - Error indication via relay contacts (potential-free) 1 NO + 1 NC contact AC15: 24 V / 4 A, 230 V / 3 A AC13: 24 V / 1 A, 220 V / 0.5 A Terminal capacity: 0.25 - 1.5 mm² - Not for single-phase use					
	400-690 V / 50-60 Hz	00	GST00...-A...	GST00-DSI 107956		1 unit
		1	GST1.-A..	GST1-DSI 107957		1 unit
	2	GST2-A...	GST2-DSI 107958		1 unit	
	3	GST3-A...	GST3-DSI 107959		1 unit	
	Terminal Capacity		Utilisation	Designation Article No.	Notes	Units per Package
	Sets of Prism Terminals					
	70-150 mm ² Cu/Al		GST1...	PSK1 038734	One set includes 3 prism terminals	1 unit
	120-240 mm ² Cu/Al		GST2...	PSK2 043480	One set includes 3 prism terminals	1 unit
	120-300 mm ² Cu/Al		GST3...	PSK3 048226	One set includes 3 prism terminals	1 unit
	Sets of Double-Prism Terminals					
	2x(70-95) mm ² Cu/Al		GST1...	PSK12 041107	One set includes 3 double- prism terminals	1 unit
	2x(120-150) mm ² Cu/Al		GST2...	PSK22 045853	One set includes 3 double- prism terminals	1 unit
	2x(120-240) mm ² Cu/Al		GST3...	PSK32 050599	One set includes 3 double- prism terminals	1 unit
	Clamp-Type Terminals					
	25-150 mm ² Cu-Band 6 x 16 x 0.8 mm		GST1...	SK1-GS 107960	You need 3 units per GST...	3 units
	25-240 mm ² Cu-Band 10 x 16 x 0.8 mm		GST2...	SK2-GS 107961	You need 3 units per GST...	3 units
	25-300 mm ² Cu-Band 11 x 21 x 1 mm		GST3...	SK3-GS 107962	You need 3 units per GST...	3 units

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Slide Fuse Equipment, 3-pole

Max. Rated Oper- ational Current I _e (A)	Max. Fuse-link 400V (A)	Max. 690V (A)	Component Size	Utilisation	Designation Article No.	Notes	Units per Package
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NH-Vertical Fuse-Switch-Disconnecter

- Incl. cover for termination space
- Drill-free mounting
- Clamp-type terminals included in the delivery
- 60 mm distance between busbars when mounted onto busbars

Without fuse monitoring

160	160	160	00	12 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10 Double-T	NH-SLS-00/160-60 106211	Connection at the top or bottom	1 / 182
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With fuse monitoring

160	160	–	00	12 x 5/10 20 x 5/10 25 x 5/10 30 x 5/10 Double-T	NH-SLS-00/160-60-SI 106216	Connection at the top or bottom	1 / 112
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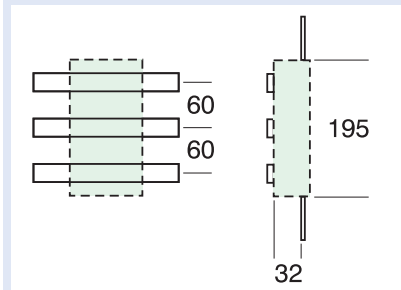
Teminal Cover/Size Compensation for GST...

For NH-SLS-00/160-60	Z-NH-SLS-KA 106223	2 units
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For NH-fuse-links Z-NH/00... and solid-links Z-NH-00/TR see chapter Accessories Fuse Devices

Coordination Table

- Combinations possible without bending the copper busbar



Device	GST-00-160-40-60-AOU	LTS-100/C00/3-R	D02-S/63/3-RS	D02-LTS/63/3-R(-HK)	D02-S0/63/3-R-27 Z-D02/R/3-..	DII-S0/25/3-R(-PS)	DIII-S0/63/3-R(-PS)
Accessory	BS-SET-GST00					SBS-RS60	SBS-RS60
Cu 12x5/10				X	X	X	X
20x5/10	X	X	X	X	X	X	X
25x5/10	X			X	X	X	X
30x5/10	X	X	X	X	X	X	X
Double-T	X	X	X	X	X	X	X

SG46912



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wa_v104807



Max. Rated Operational Current I _e (A)	Dimensions (Number of layers x width x thickness of layers) (mm)	Cross Section (mm ²)	Designation Article No.	Notes	Units per Package
Multi-layer Copper Band, insulated • E-Cu conductor, tin-plated • Rated voltage 1000 V AC • UL-listed for max. 600 V AC • Breakdown voltage 20 kV/mm • Insulating material heat resistant up to +105° Celsius • Self-extinguishing according to UL94VO • 2000 mm long					
100	3 x 9 x 0.8	21.6	CU-BAND3X9X0,8-BK 081167	black	20 units
	3 x 9 x 0.8	21.6	CU-BAND3X9X0,8-BU 080960	blue	20 units
	3 x 9 x 0.8	21.6	CU-BAND3X9X0,8-GNYE 081006	green/yellow	20 units
160	6 x 9 x 0.8	43.2	CU-BAND6X9X0,8-BK 081414	black	10 units
	6 x 9 x 0.8	43.2	CU-BAND6X9X0,8-BU 081344	blue	10 units
	6 x 9 x 0.8	43.2	CU-BAND6X9X0,8-GNYE 081367	green/yellow	10 units
200	9 x 9 x 0.8	64.8	CU-BAND9X9X0,8-BK 081515	black	10 units
	9 x 9 x 0.8	64.8	CU-BAND9X9X0,8-BU 081436	blue	10 units
	9 x 9 x 0.8	64.8	CU-BAND9X9X0,8-GNYE 081485	green/yellow	10 units
250	6 x 16 x 0.8	74.4	CU-BAND6X16X0,8-BK 081310	black	10 units
	6 x 16 x 0.8	74.4	CU-BAND6X16X0,8-BU 081222	blue	10 units
	6 x 16 x 0.8	74.4	CU-BAND6X16X0,8-GNYE 081275	green/yellow	10 units
400	10 x 16 x 0.8	124	CU-BAND10X16X0,8-BK 080739	black	5 units
	10 x 16 x 0.8	124	CU-BAND10X16X0,8-BU 079736	blue	5 units
	10 x 16 x 0.8	124	CU-BAND10X16X0,8-GNYE 080698	green/yellow	5 units
	5 x 24 x 1	120	CU-BAND5X24X1-BK 119032	black	5 units
630	11 x 21 x 1	231	CU-BAND11X21X1-BK 080923	black	5 units
	11 x 21 x 1	231	CU-BAND11X21X1-BU 080769	blue	5 units
	11 x 21 x 1	231	CU-BAND11X21X1-GNYE 080836	green/yellow	5 units
	8 x 24 x 1	192	CU-BAND8X24X1-BK 119033	black	5 units
	10 x 24 x 1	240	CU-BAND10X24X1-BK 119034	black	5 units
	5 x 32 x 1	160	CU-BAND5X32X1-BK 119035	black	5 units
800	10 x 32 x 1	320	CU-BAND10X32X1-BK 119036	black	3 units
1000	10 x 40 x 1	400	CU-BAND10X40X1-BK 119037	black	3 units
1250	10 x 50 x 1	500	CU-BAND10X50X1-BK 119038	black	2 units
1600	10 x 80 x 1	800	CU-BAND10X80X1-BK 119039	black	1 unit

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Used for	Designation Article No.	Notes	Units per Package
Line supports (Short-circuit strength diagrams see Technical Datas)			
Profile ledge	Clamp clips BZ248 076516		10 units
Clamp clips	3 x 9 x 0.8 BZ249 078889		10 units
	4 x 16 x 0.8 BZ251 081262		10 units
	6 x 16 x 0.8		
	10 x 16 x 0.8		
	11 x 21 x 1 BZ252 083635		10 units

Note

Note

This image shows a full page of blank graph paper. The background is a solid light blue color. Overlaid on this background is a precise grid of thin, dark gray horizontal and vertical lines. These lines intersect to form a series of small, identical squares that cover the entire area of the page, leaving no margins or other markings.

Table of Contents Technical Data
SASY 60i Busbar System

Current Load Busbars 60 mm System	Page 36
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Busbar supports	Page 39
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Busbar-Slide Switch-Disconnecter-Fuses	Page 46
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SASY 60i Busbar System

Technical Data on the 60 mm System

Current Load Busbars, according to DIN EN 13601

For busbar applications that have **not** been type-tested, UL508A allows an ampacity of 1000A/inch² (1.55A/mm²) if no tests have been carried out. This value may be higher if the product or the application has been tested accordingly. Moeller has **conducted extensive tests** for the user's maximum benefit in using the SASY 60i busbar system. The advantage of such tests is that one can use the SASY60i busbar system with higher rated currents than the default value allows. A busbar of size 30x10 mm for example can be charged with 630A instead of 465A only.

Higher current carrying capacities to DIN 43671 were obtained under operating conditions.

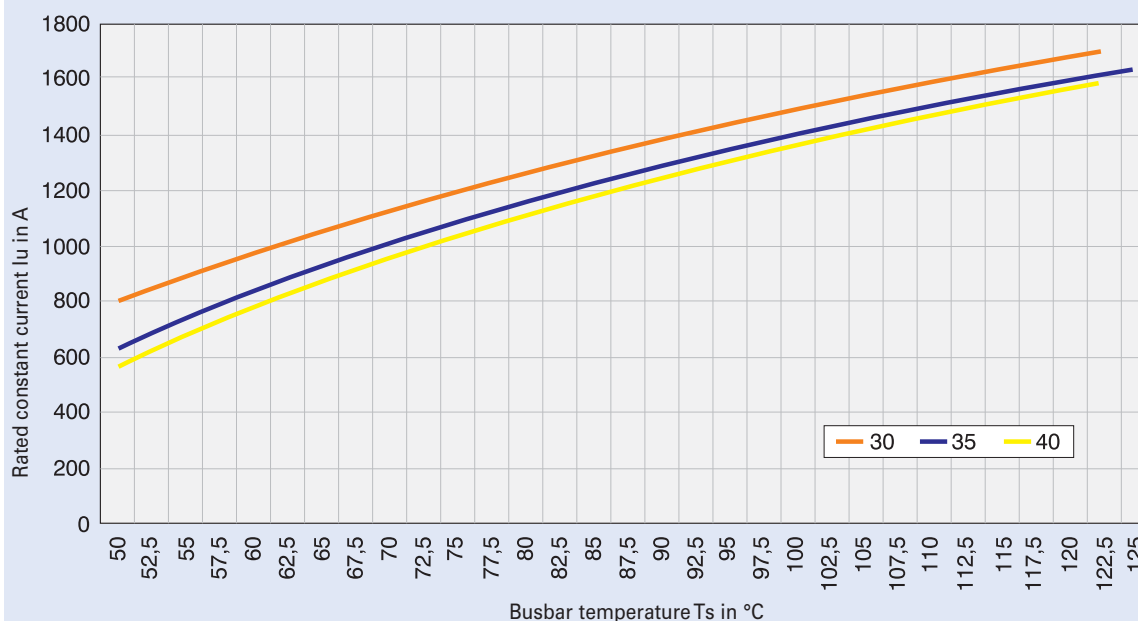
Busbar temperature is normally positively influenced by mounting components on the busbar and by air circulation within the installation.

Depending on the respective ambient temperature, you can calculate the correction factor k2 according to DIN 43 671 for flat busbars. If ambient conditions change, a correction factor needs to be taken into account.

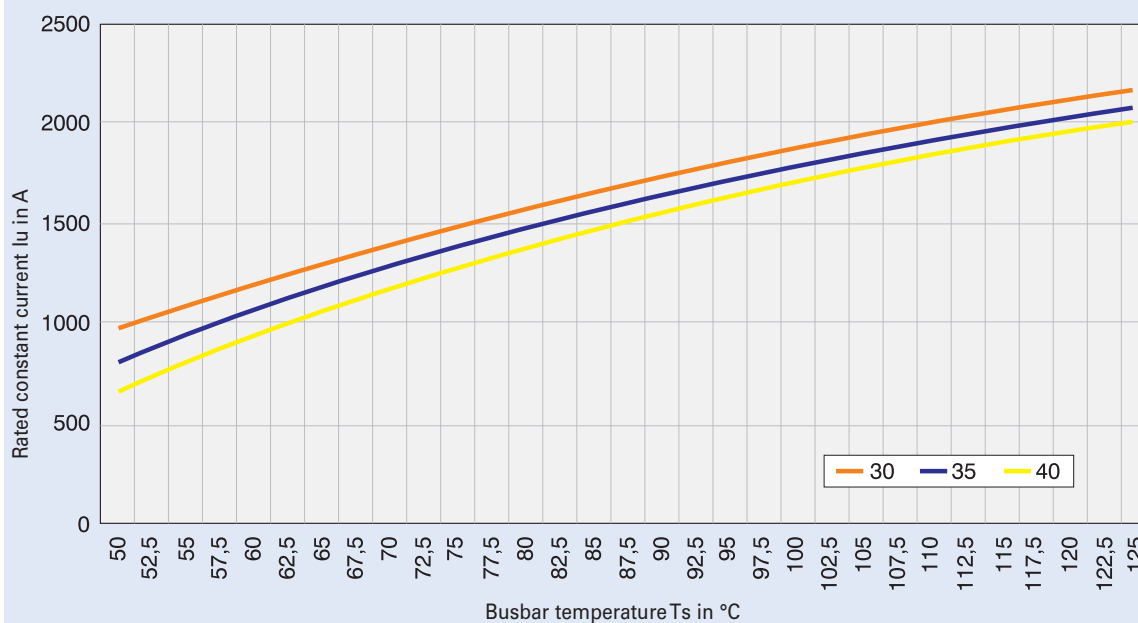
On the other hand, increased loads may occur if the components feature a correspondingly high temperature resistance.

A 30 x 10 tin-plated busbar can under normal conditions be loaded with 630 A. With a load of 800A, for instance, a k2 correction factor of 1.3 is necessary. It follows from the diagram that with this factor and 35°C air temperature, the busbar heats up to approx. 85°C.

Current load CU-BAR-500/T



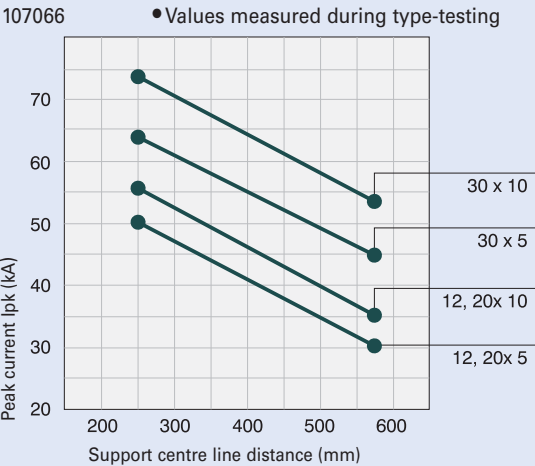
Current load CU-BAR-720/T



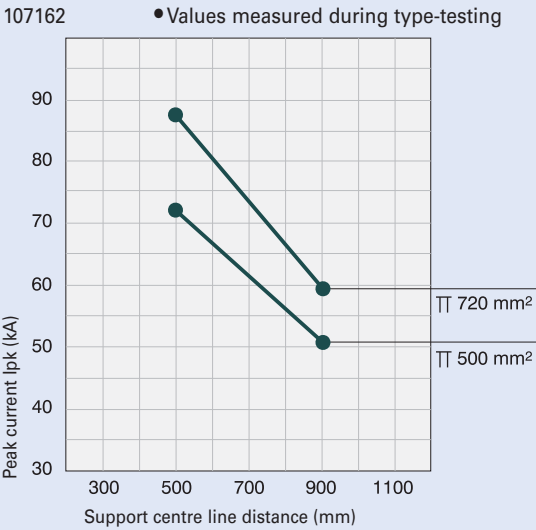
SASY 60i Busbar System

Short-circuit strength diagrams according to IEC/EN 61439-1 for 60 mm SASY 60i Busbar Systems

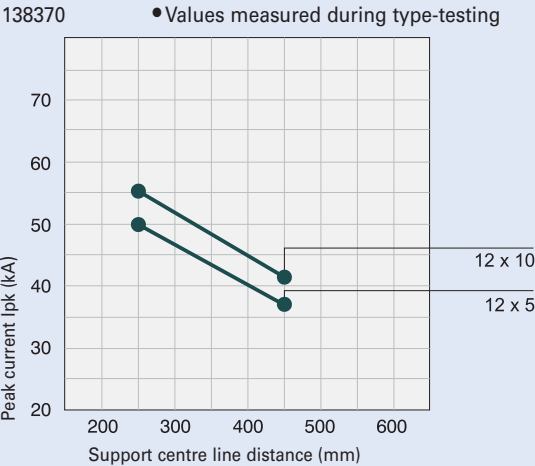
BBS-3/FL



BBS-3/PR

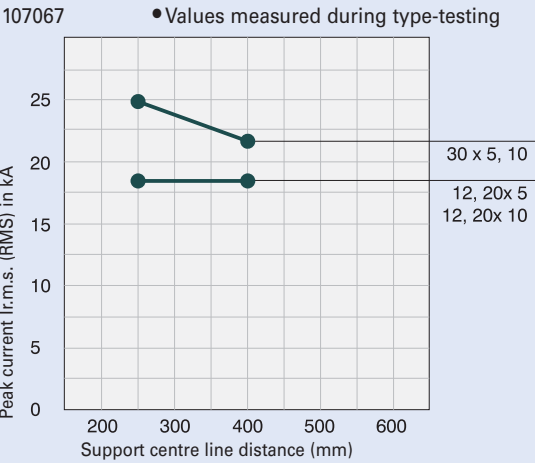


BBS-3/FL-C

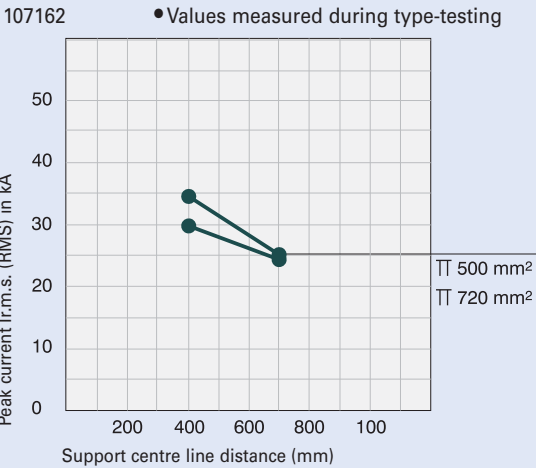


Short-circuit strength diagrams according to UL 845 for 60 mm SASY 60i Busbar Systems

BBS-3/FL-NA



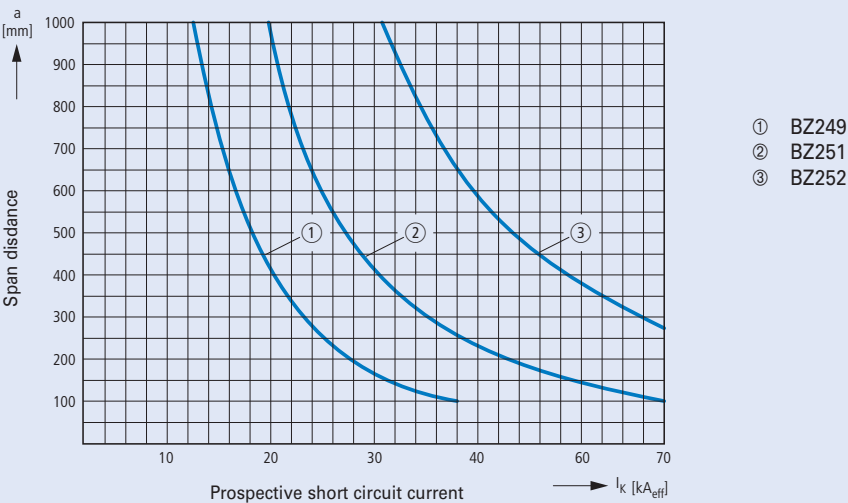
BBS-3/PR



SASY 60i Busbar System

Technical Data on Line Supports BZ

Short-circuit strength diagrams for Line Supports BZ



SASY 60i Busbar System

Technical Data on Busbar Supports

			BBS-/FL(-NA)	BBS-/PR	BBS-3/FL-C
General Information					
Standards and regulations			type-tested according to VDE 0660 Part 500 IEC/EN 61439-1		
Fitting position			vertical, horizontal		
Material					
Material			Thermoplast, silicone-free, chlorine-free		
Halogen-free			yes	yes	yes
Flammability			self-extinguishing according to UL 94		
Colour			RAL 7035	RAL 7035	RAL 7035
Comparative tracking index			CTI 200	CTI 200	CTI 200
Uninterrupted duty temperature		°C	120	120	120
Current Paths					
Rated insulation voltage		U _i	V	3000	3000
Rated operational voltage		U _e	V	690	690
Rated frequency		f	Hz	50/60	50/60
Centre line distance of busbars		mm	60	60	60
Rated uninterrupted current		In case of temperature variances, DIN 43671 requires a kA correction factor to be taken into account			
with busbar 12 x 5 mm	I _u	A	218	-	200
with busbar 15 x 5 mm	I _u	A	273	-	-
with busbar 20 x 5 mm	I _u	A	349	-	-
with busbar 25 x 5 mm	I _u	A	436	-	-
with busbar 30 x 5 mm	I _u	A	491	-	-
with busbar 12 x 10 mm	I _u	A	392	-	360
with busbar 20 x 10 mm	I _u	A	567	-	-
with busbar 30 x 10 mm	I _u	A	687	-	-
with 500 mm ²	I _u	A	-	1003	-
with 720 mm ²	I _u	A	-	1281	-
Ambient temperature		°C	35	35	35
Temperature of busbar		°C	70	70	70
Rated peak withstand current					
with busbar 12 x 5 mm	I _{pk}	kA	50	-	50
with busbar 15 x 5 mm	I _{pk}	kA	50	-	-
with busbar 20 x 5 mm	I _{pk}	kA	50	-	-
with busbar 25 x 5 mm	I _{pk}	kA	50	-	-
with busbar 30 x 5 mm	I _{pk}	kA	64	-	-
with busbar 12 x 10 mm	I _{pk}	kA	56	-	55
with busbar 20 x 10 mm	I _{pk}	kA	56	-	-
with busbar 30 x 10 mm	I _{pk}	kA	73	-	-
with 500 mm ²	I _{pk}	kA	-	72	-
with 720 mm ²	I _{pk}	kA	-	87	-
Short-circuit time		t	ms	20	20
Support centre line distance		mm	250	500	250

Technical Data on Copper Bars

			CU...	CU-BAR-...-T	CU-BAR-1140/T
General Information					
Standards and regulations			EN 13061, UL 508		
Current Paths					
Rated uninterrupted current		In case of temperature variances, DIN 43671 requires a kA correction factor to be taken into account			
T _u = 35°C and T _s = 65°C					
with busbar 12 x 5 mm	I _u	A	200	-	-
with busbar 20 x 5 mm	I _u	A	320	-	-
with busbar 30 x 5 mm	I _u	A	450	-	-
with busbar 12 x 10 mm	I _u	A	360	-	-
with busbar 20 x 10 mm	I _u	A	520	-	-
with busbar 30 x 10 mm	I _u	A	630	-	-
with 500 mm ²	I _u	A	-	950	-
with 720 mm ²	I _u	A	-	1200	-
1140 mm ²	I _u	A	-	-	2500

SASY 60i Busbar System

Technical Data on the 60 mm System

Conductor connections

The ratios between conductor cross-sections in mm² and AWG/MCM-sizes are listed below:

1.5 mm ²	16 AWG
2.5 mm ²	14 AWG
4 mm ²	12 AWG
6 mm ²	10 AWG
10 mm ²	8 AWG
16 mm ²	6 AWG
25 mm ²	4 AWG
35 mm ²	2 AWG
50 mm ²	0 AWG
70 mm ²	2/0 AWG
95 mm ²	3/0 AWG
120 mm ²	250 MCM
150 mm ²	300 MCM
185 mm ²	350 MCM
240 mm ²	500 MCM
300 mm ²	600 MCM

Busbar supports

60 mm system according to IEC

1-pole for busbars 12x5 – 30x10, double-T-bars

2-pole for busbars 12x5 – 30x10

3-pole for busbars 12x5 – 30x10 and 12/20/ 30 x 5/10

3-pole for double-T-bars

Tighten screws for fixing the cover and bottom of the support at a torque of 4 Nm min.

60 mm system according to UL

3-pole for busbars 12/20/ 30 x 5/10

3-pole for double-T-bars

Tighten screws for fixing the cover and bottom of the support at a torque of 4 Nm min.

Silicone-free, chlorine-free

Temperature resistant up to 120°C

Self-extinguishing according to UL 94

Comparative tracking index CTI 200

Busbars according to DIN EN 13601

Tin-plated Cu-bars significantly reduce the work necessary for preparing the contact points.

Cu-busbars are effectively protected against aggressive environments.

Dimension	Cross-section
Double-T	500 mm ²
Double-T	720 mm ²

Permissible tolerances:

Radius R 0.3 ... 0.7

Width: + 0.1 / – 0.5

Thickness: + 0.1 / – 0.1

Center line distance:

± 0.5 mm (60 mm system)

Variance on the contacting level: 0.4 mm

Ampacity with copper bars

Cross-sections of bars mm	Surface mm ²	Ampacity according to IEC 35°C ambient temperature 65°C bar temperature A	Ampacity according to UL/CSA A
12 x 5	60	200	200
20 x 5	100	320	320
30 x 5	150	450	450
12 x 10	120	360	360
20 x 10	200	520	520
30 x 10	300	630	630
Double-T	500	950	950
Double-T	720	1200	1200

SASY 60i Busbar System

Technical Data on the 60 mm System

Universal conductor terminal

Used for connecting conductors featuring cross-sections of 1.5–120 mm² on busbars 5 or 10 mm thick.
Integrated retaining springs, an open terminal space and captive terminal screws make the installation job easy.

Suitable conductors ¹⁾	Current carrying capacity of contact point*	Terminal space WxH mm	Busbars WxH mm	Type
1.5–16 mm ² Cu,  ,  ,  **,  8 x 6 x 0.5	180 A	7.5 x 7.5	... x 5 ... x 10	AKU16/5 AKU16/10
4–35 mm ² Cu,  ,  ,  **,  3/ 6 x 9 x 0.8	270 A	10.5 x 11	... x 5 ... x 10	AKU35/5 AKU35/10
16–70 mm ² Cu,  ,  ,  **, 2 x  3 / 6 x 9 x 0.8, 6 x 13 x 0.5	400 A	14 x 14	... x 5 ... x 10, TT	AKU70/5 AKU70/10
16–120 mm ² Cu,  ,  ,  **,  4 / 6/ 10 x 16 x 0.8	440 A	17 x 15	... x 5 ... x 10, TT	AKU120/5 AKU120/10

* Current carrying capacities specified reflect the thermal capacities of the contact points under favourable conditions (with a maximum of conductors that can be connected). They do not, however, invalidate the validity of conductor cross-sections and of current carrying capacities required by any national and international regulations.

** A reduction of maximum conductor cross-sections might be necessary.

Brace terminals

For connecting round conductors of 95–300 mm² and multi-layer copper bars.

The gripper-type of termination technology allows to embrace both sides of the busbar and to connect the conductor without drilling.

Suitable conductors ¹⁾	Current carrying capacity of contact point *	Terminal space WxH mm	Busbars WxH mm	Type
95–185 mm ² Cu, Al***  ,  , 	500 A	–	20x5 - 30x10 TT	AKS185
150–300 mm ² Cu, Al***  ,  , 	600 A	–	20x5 - 30x10 TT	AKS300
 3 x 20 x 1 to 10 x 32 x 1	800 A	32 x 25	20x5 - 30x10 TT	AKS-CU-BAND
 (2x) 50 x 10	1600 A	55 x 28	20x5 - 30x10 TT	AKS1000

*** Connections to aluminium conductors are not maintenance-free

¹⁾  Round conductor, single-wired
 Round conductor, fine-wired with expertly pressed wire end ferrule
 Round conductor, multi-wired
 Sector conductor, single-wired
 Sector conductor, multi-wired
 Cu-Band
 Cu-Bar

SASY 60i Busbar System

Technical Data on the 60 mm System

Connecting terminal plates

Incl. cover cap

16, 50, 120 mm²

3-pole, 690 V~

Centre line distance of bars 60 mm

Busbars ... x 5 – 10,

Double-T-profiles

Terminal plates:

Silicone-free, chlorine-free

Temperature resistant up to 120°C

Self-extinguishing according to UL 94

Comparative tracking index CTI 200

Cover cap:

Silicone-free, chlorine-free

Temperature resistant up to 120°C

Self-extinguishing according to UL 94

Suitable conductors ¹⁾	Current carrying capacity of contact point *	Terminal space WxH mm	Busbars WxH mm	Type
1,5 – 16 mm ² Cu,  **, 	80 A	–	... x 5 – 10 TT	BBA-TP3/16
6 – 50 (70) mm ² Cu,  **,   6 x 9 x 0.8	300 A	10 x 15	... x 5 – 10 TT	BBA-TP3/50
35 – 120 mm ² Cu,  **,   6 / 10 x 16 x 0.8	440 A	15 x 15	... x 5 – 10 TT	BBA-TP3/120

Connecting set, 3-pole

Incl. cover cap

Suitable conductors	Current carrying capacity of contact point *	Terminal space WxH mm	Busbars WxH mm	Type
120–300 mm ² Cu, Al***,  ,  , 	560 A		20x5 - 30x10 TT	BBA-TP3/300
 3 x 20 x 1 to 10 x 32 x 1 TP3/CUBAND	800 A	32 x 25	20x5 - 30x10 TT	BBA-
 , (2x) 50 x 10	1600 A	55 x 28	20x5 - 30x10 TT	AKS1000

** It might be necessary to reduce maximum conductor cross-sections.

*** Connections to aluminium conductors are not maintenance-free.

¹⁾  Round conductors, single-wired
 Round conductors, fine-wired with expertly pressed wire end ferrules
 Round conductors, multi-wired
 Sector conductors, single-wired
 Sector conductors, multi-wired
 Cu-Band
 Cu-Bar

SASY 60i Busbar System

Technical Data on the 60 mm System

Busbar connecting terminal

For lengthwise connection of identically shaped busbars without drilling

Current carrying capacity of contact point	Overall length mm	Permissible mis-alignment of bars	Spacing between systems in mm	Type
630 A	150	1 mm	100 - 110	BBT-CU12-20X5/10-150
630 A	95	5 mm	50 - 60	BBT-CU20-30X5/10-95
630 A	150	5 mm	100 - 110	BBT-CU20-30X5/10-150
1600 A	50	2 mm	9 - 20	BBT-CU-BAR500/720-50
1600 A	150	5 mm	100 - 110	BBT-CU-BAR500/720-150

Profile terminals for double-T-bars

Current carrying capacity of contact point	Profile	Terminal space W x H (without spacers)	Type
1600 A	TT	41 x 20 – 42	AKP800
1600 A	TT	51 x 20 – 42	AKP1000

Use spacers provided when two multi-layer CU-BAND types of copper busbars are connected in parallel.

xStart busbar adapter

3-pole, 690 V~

Can be used on all busbars in a 60 mm system.

Thanks to the combi-base it is suitable for a thickness of both 5 and 10 mm.

DIN EN 60715 support rail, plastic, can be adjusted on a 1.25-mm grid.

Copper conductors are ultrasound welded.

Base body:

Silicone-free, chlorine-free

Temperature resistant up to 120°C

Self-extinguishing according to UL 94

Comparative tracking index CTI 200

Support rails:

Silicone-free, chlorine-free

Temperature resistant up to 100°C

PVC conductor insulation:

Temperature resistant up to 105°C

Overall length of the connecting cables:

BBA0-25, BBA0-32, BBA0R-25, BBA0R-32, BBA0-25/2TS, BBA0/2TS-L : 93 mm

BBA0C-16, BBA0RC-16: 125 mm

BBA4-63, BBA2-63, BBA4L-63, BBA2L-63: 115 mm

Short circuit current ratings SCCR:

PKZM0-0,16 to PKZM0-10: 50 kA

PKZM0-12, -16: 50 kA

PKZM0-20, -25, -32: 18 kA

Direct and reverse motor starters (PKZM0 + DILM 1) or MSC + BBA up to 32 A on busbar adapters and busbar system

SASY 60i

¹⁾ any type of connection between PKZM and DILM

SASY 60i Busbar System

Technical Data on the 60 mm System

Busbar adapter NZM

Parameter	NZM1-XAD160	NZM2-XAD250	NZM3-XAD630
Design	3-pole, 690 V~	3-pole, 690 V~	3-pole, 690 V~
Bar system	60 mm	60 mm	60 mm
Bar contacting	combi-base	claw-type terminal	claw-type terminal
Connection of the switchgear	top/bottom	top or bottom	top or bottom
Short Circuit Current Ratings SCCR	32 kA at 480 V	35 kA at 480 V 50 kA at 600 V	65 kA at 480 V 50 kA at 600 V

NZM1-XAD160

Base body:

Thermoplast

Temperature resistant up to 120°C

Self-extinguishing according to UL 94

Comparative tracking index CTI 200

Halogen-free

Derating:

Ambient temperature [°C]	25	30	35	40	45	50	55
Permissible rated current [A]	160	155	150	146	141	136	130
Derating to 160 A	1	0.97	0.94	0.91	0.88	0.85	0.81

NZM2-XAD250

Base body:

Thermoplast

Temperature resistant up to 120°C,

Self-extinguishing according to UL 94,

Comparative tracking index CTI 200,

Halogen-free

NZM3-XAD630

Base body:

Thermoplast

Temperature resistant up to 120°C,

Self-extinguishing according to UL 94,

Comparative tracking index CTI 200,

Halogen-free

Derating:

Ambient temperature [°C]	20	30	40	50	60	65	70
Permissible rated current [A]	630	605	580	554	529	517	504
Derating to 630 A	1	0.96	0.92	0.88	0.84	0.82	0.80

Note

Please observe the de-rating coefficients listed in the table above to determine the maximum ampacity allowed at different ambient temperatures!

Example

An NZM3...3-...630... device with an NZM3-XAD630 device adapter should be operated at an ambient temperature of 50°C.

Question

What is the maximum rated operating current I_e allowed I_e ? =>

Solution

At an ambient temperature of 50°C, the de-rating coefficient is 0.88. This means that $I_e = 630A \times 0,88 = 544A$.
Der Schalter darf also bei 50°C, the device can therefore be operated at a maximum of $I_e = 544 A$.

SASY 60i Busbar System

D-Type Slide Fuse-Base

- Design according to IEC/EN 60269-1, VDE 0636 Part 301
- Vertical and horizontal mounting possible
- Delivered empty, without screw caps

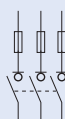
Technical Data

	D02-SO/63/3-R-27 Z-D02/R/3...	DII-SO/25/3-R(-PS)	DIII-SO/63/3-R(-PS)
Electrical			
Number of poles	3	3	3
Rated operational voltage U_e	400 V AC	500 V AC	690 V AC
Rated frequency	40-60 Hz	40-60 Hz	40-60 Hz
Rated operational current I_e	63 A	25 A	63 A
Conv. thermal current with fuse-links I_{th}	63 A	25 A	63 A
Rated duty	uninterrupted duty	uninterrupted duty	uninterrupted duty
Rated conditional short-circuit current	50 kA _{r.m.s.}	50 kA _{r.m.s.}	50 kA _{r.m.s.}
Overvoltage category	IV	III	III
Rated impulse withstand voltage U_{imp}	6 kV	4 kV	4 kV
Power loss per current path	0.5 W	0.4 W	3.34 W
Power loss of base without fuse-links	1.5 W	1.2 W	10 W
Max. permissible power loss of fuse-links	5.5 W	4 W	7 W
Mechanical			
Device height	201 mm	200 mm	200 mm
Width	27 mm	45 mm	54 mm
Weight	150 g	140 g	150 g
Mounting onto busbars, without drilling or screwing	12x5/10 15x5/10 20x5/10 25x5/10 30x5/10	12x5/10 - 20x5/10 25x5/10 30x5/10	12x5/10 - 20x5/10 25x5/10 30x5/10
Degree of protection while operating	IP20	IP20	IP20
Terminals	Lift terminals	Lift terminals	Lift terminals
Terminal capacity	1.5-35 mm ²	1.5-25 mm ²	1.5-25 mm ²
Tightening torque of terminal screws	3-4 Nm	2.6 Nm	2.6 Nm
Electrical thread type	E18	E27	E33
Ambient temperature range	-25 to +55°C	-25 to +55°C *)	-25 to +55°C *)
*) (35°C normal temperature, at 55°C with reduced operating current)			
Pollution degree	3	3	3
Climatic resistance: moist heat	constant acc. to IEC 60068-2-78, cyclical acc. to IEC 60068-2-30		

Busbar-Slide Switch Disconnecter with Fuses D02-S/63/3-RS

- Design according to IEC/EN 60947-3
- Vertical and horizontal mounting possible
- Supplied empty, without screw caps
- Current coding by means of cartridge-ring adapter insert
- Suitable for fuse-links
 - D01: 2, 4, 6, 10, 16 A in combination with cartridge-ring adapter inserts Z-D02-D01/PE-.. and adapter spring Z-D02/SIKA-HF
 - D02: 20, 25, 35, 50, 63 A
- Can be sealed with lead

Connection diagram



Technical Data

Electrical		Mechanical	
Number of poles	3P	Device height	212 mm
Rated operational voltage U_e		Width	36 mm
AC	400 V / 40-60 Hz	Weight	260 g
Rated operational current I_e	63 A	Mounting onto busbars, without drilling or screwing	20x5/10 mm 30x5/10 mm
Conv. thermal current with fuse-links I_{th}	63 A	Degree of protection while operating	IP30
Rated duty	Uninterrupted duty	Terminals	Lift terminals
Rated conditional short-circuit current	50 kA _{r.m.s.}	Terminal capacity	1.5-25 mm ² Cu
Utilization category	AC 23 B	Tightening torque of terminal screws	max. 2.6 Nm
Overvoltage category	III	Electrical thread type	E18
Rated impulse withstand voltage U_{imp}	8 kV	Temperature range	-25 to +55°C
Power loss per current path	2 W with I_e	Pollution degree	3
Power loss per current path with fuse-link	7.5 W with I_e	Climatic resistance: moist heat	constant acc. to IEC 60068-2-78, cyclical acc. to IEC 60068-2-30
Max. permissible power loss of fuse-links	5.5 W		

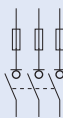
SASY 60i Busbar System

Busbar-Slide Switch-Disconnecter-Fuse D02-LTS/63/3-R(-HK)

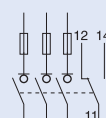
- Design according to IEC/EN 60947-3
- Vertical and horizontal mounting possible
- Supplied empty
- Current coding by means of cartridge-ring adapter insert
- Suitable for fuse-links
 - D01: 2, 4, 6, 10, 16 A in combination with cartridge-ring adapter inserts Z-D02-D01/PE-.. and adapter spring Z-D02-LTS-HF
 - D02: 20, 25, 35, 50, 63 A
 - Cylindrical 10x38: 1 - 32 A
- Can be sealed with leads, lockable

Connection diagram

D02-LTS/63/3-R



D02-LTS/63/3-R-HK



Technical Data

Electrical

Number of poles	3P
Rated operational voltage U_e	
AC	400 V / 40-60 Hz
Rated operational current I_e	63 A
Rated uninterrupted current I_u	63 A
Rated duty	Uninterrupted duty
Rated short-circuit capacity I_{cm}, I_{cn}	50 kA _{r.m.s.}
Utilization category	AC 22 B
Overvoltage category	IV
Rated impulse withstand voltage U_{imp}	6 kV
Power loss per current path	1.5 W with I_e
Power loss per current path with fuse-link	7 W with I_e
Max. permissible power loss of fuse-links	5.5 W

Mechanical

Device height	226 mm
Width	27 mm
Weight	340 g
Mounting onto busbars, without drilling or screwing	12x5/10 mm
	15x5/10 mm
	20x5/10 mm
	25x5/10 mm
	30x5/10 mm
Degree of protection while operating (built-in)	IP20/IP40
Terminals	Lift terminals
Terminal capacity	1.5-35 mm ² Cu
Tightening torque of terminal screws	max. 4 Nm
Temperature range	-25 to +55°C
Pollution degree	3
Climatic resistance: moist heat	constant acc. to IEC 60068-2-78, cyclical acc. to IEC 60068-2-30

Auxiliary switch electrical

1 CO	5 A / 250 V AC
Max. thermal back-up fuse	2 A gL
	PLSM-B4/...HS / CLS6-B4/...HS

Connection

Femal push-on connector	2.8 x 0.5 mm
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SASY 60i Busbar System

NH-Fuse-Switch-Disconnecter LTS-100/C00/3-R, GST00-160-40-60-AOU..

- Mounting height max. 2000 mm
- Break-proof, flexible connection chamber
- Fully insulated, shock hazard protected in acc. with IEC/EN 60947 or BGV A3
- The base body is made of glass fibre reinforced, high-temperature resistant, self-extinguishing and halogen-free plastic
- The contact system consists of separately spring-loaded silver-plated copper contacts
- The switch cover is made of glass fibre reinforced, self-extinguishing, silicone and chlorine-free thermoplast
- The switch cover features large viewing windows allowing to read the labelling and to see the flat indicator of the NH-fuse-links
- The viewing windows provide test holes
- Vertical and horizontal mounting position is possible
- Delivery without NH-fuse-links

Connection diagram



		LTS-100/C00/3-R	GST00-...-AOU	GST00-...-AOU-F
Electrical				
Technical data according to		IEC/EN 60947-3		
Rated operational voltage	Ue V AC	500	500 / 690	500 / 690
Rated operational voltage	Ue V DC	220	220 / 440	220 / 440
Rated operational current	Ie A	100	160 / 100	160 / 100
Rated frequency	Hz	40 - 60	40 - 60	40 - 60
Rated conditional short-circuit current AC	kAr.m.s.	50	50	50
Rated conditional short-circuit current DC	kAr.m.s.	25	25	25
Utilization category AC 22 B				
Rated making capacity	A	300	480 / 300	480 / 300
Rated breaking capacity	A	300	480 / 300	480 / 300
Utilization category DC 21 B				
Rated making capacity	A	400	150	150
Rated breaking capacity	A	400	150	150
Service life electrical - operating cycles		300	300	300
Service life mechanical - operating cycles		1700	1700	1700
Power loss with Ith AC, without NH-SE	W	11.5	6.9 / 2.7	6.9 / 2.7
Power loss with Ith DC, without NH-SE	W	7.7	4.6 / 1.8	4.6 / 1.8
Rated insulation voltage	Ui V AC	500	750	750
Overvoltage category		III	III	III
Rated impulse withstand voltage	U _{imp} kV	8	8	8
Max. Fuse-link				
Size		NH000	NH00	NH00
Max. rated current gL/gG	A	100	160	160
Max. permissible power loss of NH-SE	Pv W	7.5	12	12
Mechanical				
Busbar mounting	mm	20x5/10 30x5/10	20x5/10 25x5/10 30x5/10	20x5/10 25x5/10 30x5/10
Tightening torque	Nm	–	4	4
Ambient temperature range	°C	-25 to +55	-25 to +55	-25 to +55
Degree of protection (open front cover)		IP20 (IP10)	IP20 (IP10)	IP20 (IP10)
Pollution degree		3	3	3
Weight	kg	0.57	0.93	0.93
Climatic resistance: moist heat				
constant according to IEC 60068-2-78, cyclical according to IEC 60068-2-30				
Cross sections of connections				
Flat connection (F)				F ¹⁾
Screw				M8
Cable lug	mm ²			1 x 10-95
Flat busbar	mm			20 x 10
Tightening torque	Nm			12 - 15
Clamp-type terminal (S) / Lift terminal (K)		K ¹⁾	K ¹⁾	S
Stranded Cu	mm ²	1.5 - 50	1.5 - 70	10 - 95
Cu-Band, No. of layers	mm	6 x 9 x 0.8	6 x 9 x 0.8	6 x 9 x 0.8
	x width x thickness			
Tightening torque	Nm	2.6	2.6	2.6
Prism terminal				
Stranded Al/Cu	mm ²			10 - 70
Tightening torque	Nm			2.6
Double-prism terminal				
Stranded Al/Cu	mm ²			
Tightening torque	Nm			

¹⁾ Standard-connection in the delivery condition

SASY 60i Busbar System

NH-Fuse-Switch-Disconnecter GST...

- Mounting height max. 2000 m
- Break-proof, flexible connection chamber
- Fully insulated, shock hazard protected in acc. with IEC/EN 60947 or BGV A3
- The base body is made of glass fibre reinforced, high-temperature resistant, self-extinguishing and halogen-free plastic
- The contact system consists of separately spring-loaded silver-plated copper contacts
- The switch cover is made of glass fibre reinforced, self-extinguishing, silicone and chlorine-free thermoplast
- The switch cover features large viewing windows allowing to read the labelling and to see the flat indicator of the NH-fuse-links
- The viewing windows provide test holes
- Vertical and horizontal mounting position is possible
- Delivery without NH-fuse-links

Connection diagram



			GST1..	GST2..	GST3..
Electrical					
Technical data according to			IEC/EN 60947-3		
Rated operational voltage	Ue	V AC	500 / 690	500 / 690	500 / 690
Rated operational voltage	Ue	V DC	220 / 440	220 / 440	220 / 440
Rated operational current	Ie	A	250 / 200	400 / 315	630 / 500
Rated frequency		Hz	40 - 60	40 - 60	40 - 60
Rated conditional short-circuit current AC		kAr.m.s.	50	50	50
Rated conditional short-circuit current DC		kAr.m.s.	25	25	25
Utilization category AC 22 B					
Rated making capacity		A	750 / 600	1200 / 945	1890 / 1500
Rated breaking capacity		A	750 / 600	1200 / 945	1890 / 1500
Utilization category DC 21 B					
Rated making capacity		A	300	475	750
Rated breaking capacity		A	300	475	750
Service life electrical - operating cycles			200	200	200
Service life mechanical - operating cycles			1400	800	800
Power loss with Ith AC, without NH-SE		W	12.9 / 8.3	27 / 16.7	52 / 32.8
Power loss with Ith DC, without NH-SE		W	8.6 / 5.5	18 / 11.2	34.7 / 21.8
Rated insulation voltage	Ui	V AC	750	750	750
Overvoltage category			III	III	III
Rated impulse withstand voltage	U _{imp}	kV	8	8	8
Max. Fuse-link					
Size			NH1	NH2	NH3
Max. rated current gL/gG		A	250	400	630
Max. permissible power loss of NH-SE	Pv	W	23	34	48
Mechanical					
Busbar mounting		mm	20x5/10 30x5/10	20x5/10 30x5/10	20x5/10 30x5/10
Tightening torque		Nm	6	6	6
Ambient temperature range		°C	-25 to +55	-25 to +55	-25 to +55
Degree of protection (open front cover)			IP20 (IP10)	IP20 (IP10)	IP20 (IP10)
Pollution degree			3	3	3
Weight		kg	4.4	5.3	6.6
Climatic resistance: moist heat			constant according to IEC 60068-2-78, cyclical according to IEC 60068-2-30		
Cross sections of connections					
Flat connection (F)			F ¹⁾	F ¹⁾	F ¹⁾
Screw			M10	M10	M10
Cable lug		mm ²	1 x 25-150	1 x 25-240	1 x 25-300
Flat busbar		mm	30 x 10	30 x 10	30 x 10
Tightening torque		Nm	30 - 35	30 - 35	30 - 35
Clamp-type terminal (S) / Lift terminal (K)			S	S	S
Stranded Cu		mm ²	25 - 150	25 - 240	25 - 300
Cu-Band, No. of layers		mm	6 x 16 x 0.8	10 x 16 x 0.8	11 x 21 x 1
		x width x thickness			
Tightening torque		Nm	9.5	23	23
Prism terminal					
Stranded Al/Cu		mm ²	70 - 150	120 - 240	120 - 300
Tightening torque		Nm	4.5	11	11
Double-prism terminal					
Stranded Al/Cu		mm ²	2x70-95	2x120-150	2x120-240
Tightening torque		Nm	4.5	11	11

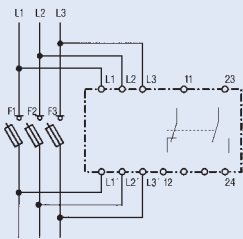
¹⁾ Standard-connection in the delivery condition

SASY 60i Busbar System

Switch Cover with Fuse Monitoring for NH-Fuse-Switch-Disconnecter GST...

- Operation indication
In working order -> 1 green LED
Fuse blown -> 3 red LEDs L1, L2, L3
- Working current principle
- VDE-requirements regarding touch voltage (> 1000 Ohm/V) are complied with.
To release, turn off the upstream main switch!
- Not applicable for single phase application

Connection diagram



Technical data

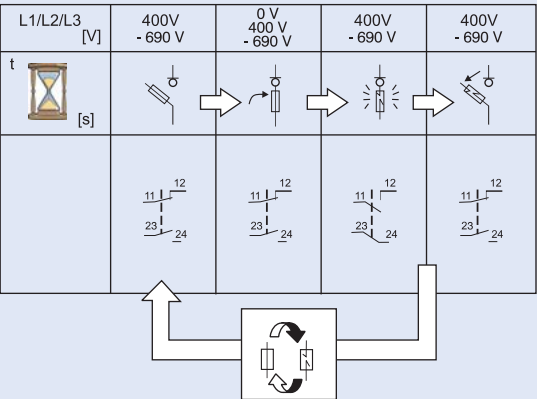
Electrical

Technical data according to	IEC/EN 60255, VDE 0435 Part 303
Rated voltage U_e	3 x 400 V AC - 690 V AC
Voltage range	0,9 ... 1,1x U_e
Rated frequency	50-60 Hz
Consumption	1.5 VA (L2/L3)
Rated insulation voltage U_i	690 V
Overvoltage category	III
Rated impulse withstand voltage U_{imp}	6 kV
Relay contacts	1 NC, 1 NO
Technical data according to	EN 60204, EN 50178, VDE 0106
Rated voltage	250 V AC
Rated current I_{th}	8 A DC
Max. making current	15 A
Utilization category	AC 15 DC13
Rated operational voltage U_e	230VAC 24VAC 24VDC 220VDC
Rated operational current I_e	1 A 4 A 1 A 0,5 A
Minimum switching load	5 V AC/DC, 300 mW, 5 mA
Max. switching capacity	
AC 1	2000 VA
250 V AC 15	400 VA
250 V AC 3	300 W
Endurance electrical comp.	80 x 10 ³ operating cycles AC 1
Maximum back-up fuse	4 A gL/gG / PLSM-B4...-HS

Mechanical

Terminals	Lift terminals
Terminal capacity stranded	0.25 - 1.5 mm ²
Tightening torque of terminal screws	0.5 - 0.6 Nm
Endurance mechanical comp.	> 10 x 10 ⁶ operating cycles
Response/Release time	< 500 ms
Depree of protection	IP20
Temperature range	-10 to +55°C
Pollution degree	3

Functional diagram

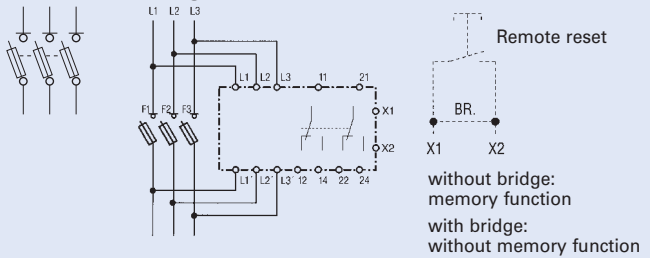


SASY 60i Busbar System

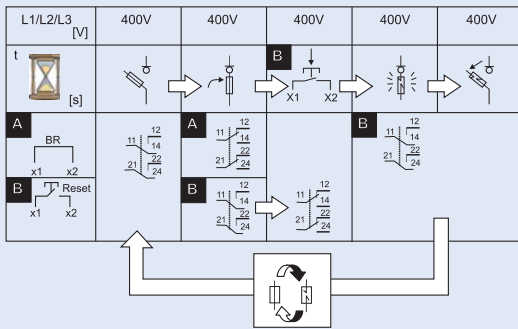
NH-Vertical Fuse-Switch-Disconnecter NH-SLS, 3-pole

- Supplied without NH-fuse-links
- Can be installed both in a vertical and horizontal position
- Symmetrical device - top or bottom connection is possible
- Break-proof, flexible connection chamber
- Fully insulated, shock hazard protected in accordance with IEC/EN 60947 or BGV A3
- The base body is made of glass fibre reinforced, high-temperature resistant, self-extinguishing and halogen-free plastic
- The contact system consists of separately spring-loaded silver-plated copper contacts
- The switch cover is made of glass fibre reinforced, self-extinguishing, silicone and chlorine-free thermoplast
- The switch cover features large viewing windows allowing to read the labelling and to see the flat indicator of the NH-fuse-links
- The viewing windows provide self-closing test holes
- The switch cover can be deposited ("park position")

Connection diagram



Functional diagram



Electronic Fuse Monitoring

- Operation indication
Fuse in working order → 1 green LED
Fuse blown, cover open → 1 red LED
- Working current principle, alternatively with memory function/remote reset
- Internal resistance of the measuring paths within the MOhm-range, VDE-requirements regarding touch voltage (> 1000 Ohm/V) are complied with. To release, turn off the upstream main switch!
- When using feeder bars the lines connected to the fuse monitoring system must be changed as follows: L1↔L1', L2↔L2', L3↔L3'

Technical Data

NH-SLS-00/160-60(-SI)		Electronic Fuse Monitoring	
Electrical		Electrical	
Technical data according to	IEC/EN 60947-3	Technical data according to	IEC/EN 60255, VDE 0435 Part 303
Size	00	Rated voltage U_e	3 x 400 V AC
Number of poles/phases	3	Voltage range	0,8 ... 1,1x U_e
Conventional free air thermal current I_{th} with NH-fuse-links	160 A	Rated frequency	50-60 Hz
Max. permissible power loss of NH-fuse-links	12 W	Consumption	2 VA (L2/L3)
Utilization category AC 23 B		Rated insulation voltage U_i	400 V
Rated operational voltage U_e	400 V AC	Overvoltage category	III
Rated operational current I_e	160 A	Rated impulse withstand voltage U_{imp}	4 kV
Rated conditional short-circuit current with fuse-links	50 kA	Relay contacts	2 CO
Utilization categories AC 22 B, AC 21 B		Technical data according to	EN 60947-5-1
Rated operational voltage U_e	690 V AC (400 V AC)	Rated voltage	250 V AC
Rated operational current I_e	160 A	Rated current I_{th}	4 A
Rated conditional short-circuit current with fuse-links	50 kA	Utilization category	AC 15
Rated insulation voltage U_i	1000 V (400 V)	Rated operational voltage U_e	230 V AC
Overvoltage category	III	Rated operational current I_e	1 A
Rated impulse withstand voltage U_{imp}	8 kV (4 kV)	Endurance electrical comp.	1.5 x 10 ⁵ operating cycles
Rated frequency	50-60 Hz	Maximum back-up fuse	4 A gL/gG / PLSM-B4...-HS
Rated duty	uninterrupted duty *		
Power loss without NH-fuse-links	20 W at 160A		
Mechanical		Mechanical	
Mounting onto busbar system without drilling or screwing		Terminals	Lift terminals
12x5/10 mm		Terminal capacity	
20x5/10 mm		solid	2 x 2.5 mm ²
25x5/10 mm		stranded	2 x 1.5 mm ²
30x5/10 mm		Tightening torque	
Standard connection - screw	M8	of terminal screws	0.8 Nm
Cable lugs - Cu-conductor according to rated currents as detailed in IEC/EN 60947-1	1 x 70 mm ²	Endurance mechanical comp.	> 10 ⁸ operating cycles
Flat busbars	20x8 mm	Response/Release time	< 500 ms
Tightening torque	12 Nm		
Standard connection - clamp strap	2 x M5		
Stranded Cu	1.5 x 70 mm ²		
Copper band: No. of layers x width x height	9 x 9 x 0.8 mm		
Flat busbars	12x10 mm		
Tightening torque	3 Nm		
Ambient temperature range	-5°C to +40°C		
Degree of protection	IP30		
Pollution degree	3		

* In case of uninterrupted duty of several devices placed side by side, please take account of the rated load factor detailed in VDE 0660 Part 500 / EN 61439-1, Table 1.

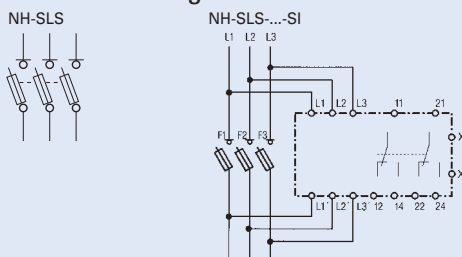
For sizes 00/1-3 the distance to grounded elements must be at least 50/100 mm at the top and 25/50 mm at the side.

SASY 60i Busbar System

NH-Vertical Fuse-Switch-Disconnecter NH-SLS, 3-pole

- Supplied without NH-fuse-links
- Can be installed both in a vertical and horizontal position
- Symmetrical device - top or bottom connection is possible
- Break-proof, flexible connection chamber
- Fully insulated, finger and hand touch-safe in accordance with IEC/EN 60947 or BGV A3
- The base body is made of glass fibre-reinforced, high-temperature resistant, self-extinguishing and halogen-free plastic
- The contact system consists of separately spring-loaded silver-plated copper contacts
- The switch cover is made of glass fibre-reinforced, self-extinguishing, silicone and chlorine-free thermoplast
- The switch cover features large viewing windows allowing to read the labelling and to see the flat indicator of the NH-fuse-links
- The viewing windows provide self-closing test holes
- The switch cover can be deposited ("park position")

Connection diagram



Technical Data

	NH-SLS-00/160(-SI)	NH-SLS-1/250(-SI)	NH-SLS-2/400(-SI)	NH-SLS-3/630(-SI)
Electrical				
Technical data according to	IEC/EN 60947-3	IEC/EN 60947-3	IEC/EN 60947-3	IEC/EN 60947-3
Size	00	1	2	3
Number of poles/phases	3	3	3	3
Conventional free air thermal current I_{th} with NH-fuse-links	160 A	250 A	400 A	630 A
Max. permissible power loss of NH-fuse-links	12 W	23 W	34 W	48 W
Utilization category AC 23 B				
Rated operational voltage U_e	400 V AC	500 (400) V AC	500 (400) V AC	500 (400) V AC
Rated operational current I_e	160 A	250 A	400 A	630 A
Rated conditional short-circuit current with fuse-links	50 kA	100 kA	100 kA	100 kA
Utilization categories AC 22 B, AC 21 B				
Rated operational voltage U_e	690 (-) V AC	690 (-) V AC	690 (-) V AC	690 (-) V AC
Rated operational current I_e	160 (-) A	250 (-) A	400 (-) A	630 (-) A
Rated conditional short-circuit current with fuse-links	50 kA	100 kA	100 kA	100 kA
Rated insulation voltage U_i	1000 V	1000 V	1000 V	1000 V
Overvoltage category	III	III	III	III
Rated impulse withstand voltage U_{imp}	8 (4) kV	12 (4) kV	12 (4) kV	12 (4) kV
Rated frequency	50-60 Hz	50-60 Hz	50-60 Hz	50-60 Hz
Rated duty	uninterrupted duty *	uninterrupted duty *	uninterrupted duty *	uninterrupted duty *
Power loss without NH-fuse-links	20 W at 160A	30 W at 250A	67 W at 400A	116 W at 630A
Mechanical				
Mounting onto busbar system				
Terminal clamps for busbar thickness	10 mm	10 mm	10 mm	10 mm
Screws for busbar thickness	5-10 mm	5-10 mm	5-10 mm	5-10 mm
	M8	M12	M12	M12
Standard connection - screw	M8	M10	M10	M12
Cable lugs - Cu-conductor according to rated currents as detailed in IEC/EN 60947-1	1 x 70 mm ²	120 mm ²	240 mm ²	2 x 185 mm ²
Busbars of a max. width of	20 mm	30 mm	30 mm	30 mm
Ambient temperature range	-5°C to +40°C	-5°C to +40°C	-5°C to +40°C	-5°C to +40°C
Degree of protection	IP30	IP30	IP30	IP30
Pollution degree	3	3	3	3

* In case of uninterrupted duty of several devices placed side by side, please take account of the rated load factor detailed in VDE 0660 Part 500 / EN 61439-1, Table 1.

For sizes 00/1-3 the distance to grounded elements must be at least 50/100 mm at the top and 25/50 mm at the side.

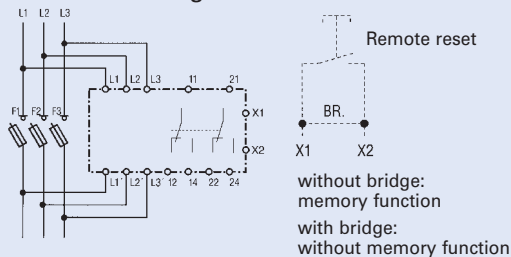
SASY 60i Busbar System

NH-Vertical Fuse-Switch-Disconnecter NH-SLS-...-SI, 3-pole

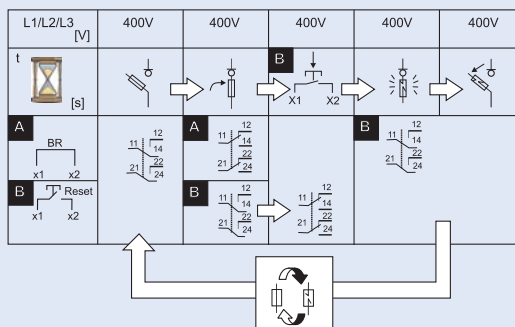
Electronic Fuse Monitoring

- Operation indication
Fuse in working order → 1 green LED
Fuse blown, cover open → 1 red LED
- Working current principle, alternatively with memory function/remote reset
- Internal resistance of the measuring paths within the MOhm-range, VDE-requirements regarding touch voltage (> 1000 Ohm/V) are complied with.
To release, turn off the upstream main switch!
- When using feeder bars the lines connected to the fuse monitoring system must be changed as follows: L1↔L1', L2↔L2', L3↔L3'

Connection diagram



Functional diagram



Technical Data

Electrical

Technical data according to	IEC/EN 60255, VDE 0435 Part 303
Rated voltage U_e	3 x 400 V AC
Voltage range	0,8 ... 1,1x U_e
Rated frequency	50-60 Hz
Consumption	2 VA (L2/L3)
Rated insulation voltage U_i	400 V
Overvoltage category	III
Rated impulse withstand voltage U_{imp}	4 kV

Relay contacts	2 CO
Technical data according to	EN 60947-5-1
Rated voltage	250 V AC
Rated current I_{th}	4 A
Utilization category	AC 15
Rated operational voltage U_e	230 V AC
Rated operational current I_e	1 A
Endurance electrical comp.	1.5 x 10 ⁵ operating cycles
Maximum back-up fuse	4 A gL/gG / PLSM-B4...-HS

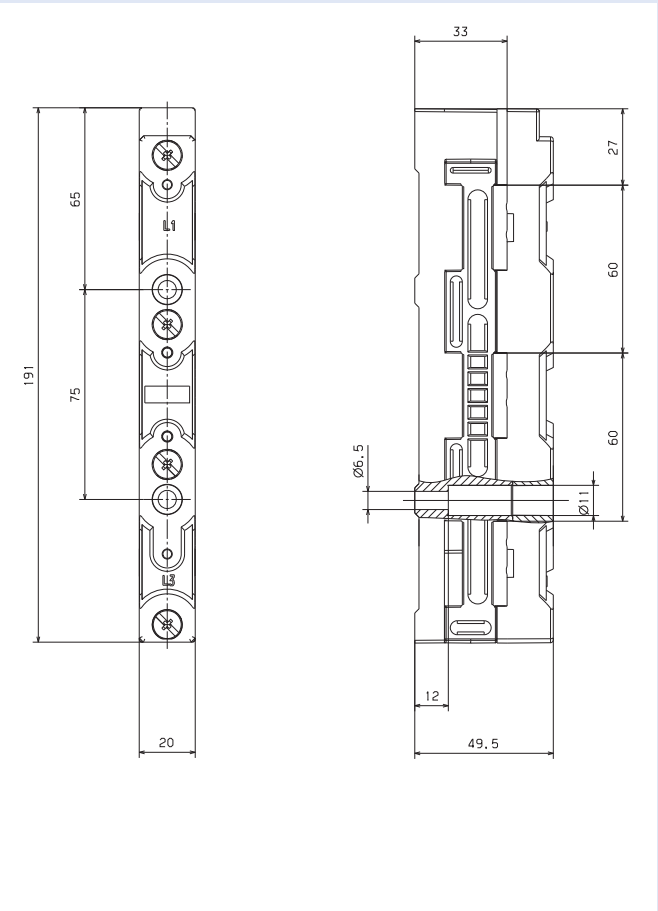
Mechanical

Terminals	Lift terminals
Terminal capacity	
solid	2 x 2.5 mm ²
stranded	2 x 1.5 mm ²
Tightening torque of terminal screws	0.8 Nm
Endurance mechanical comp.	> 10 ⁸ operating cycles
Response/Release time	< 500 ms

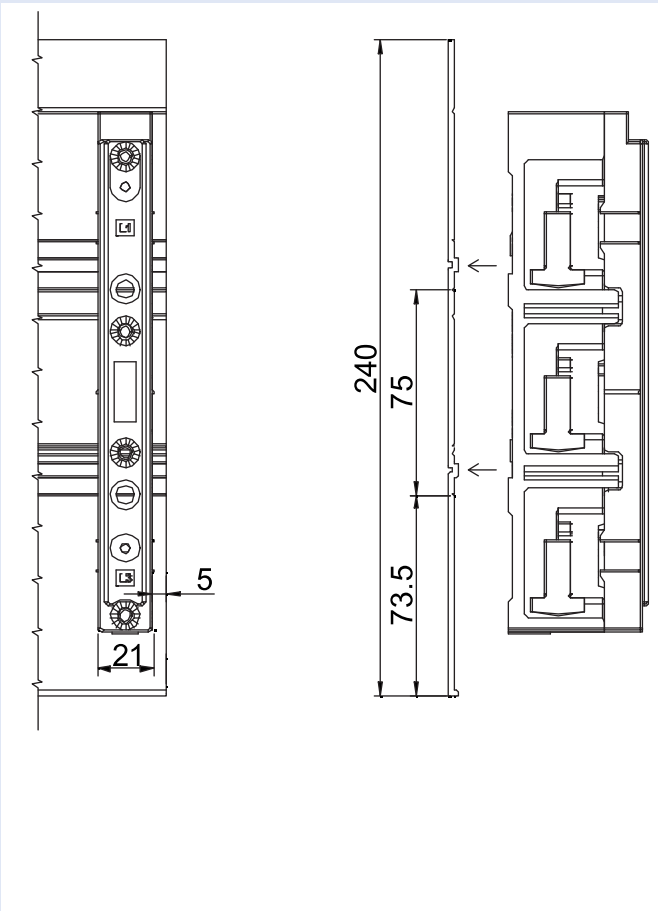
SASY 60i Busbar System

Dimensions

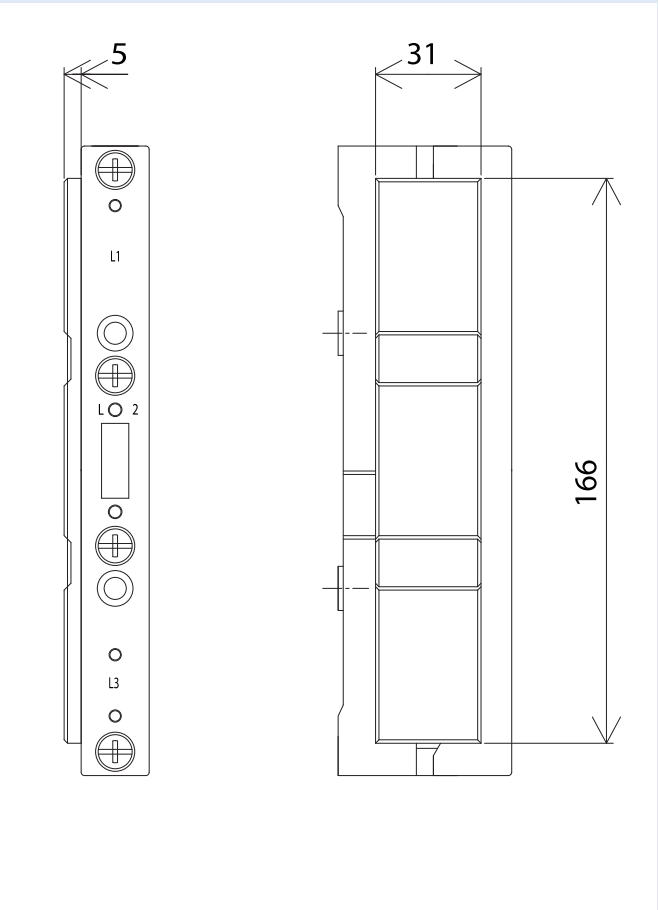
BBS-3/FL



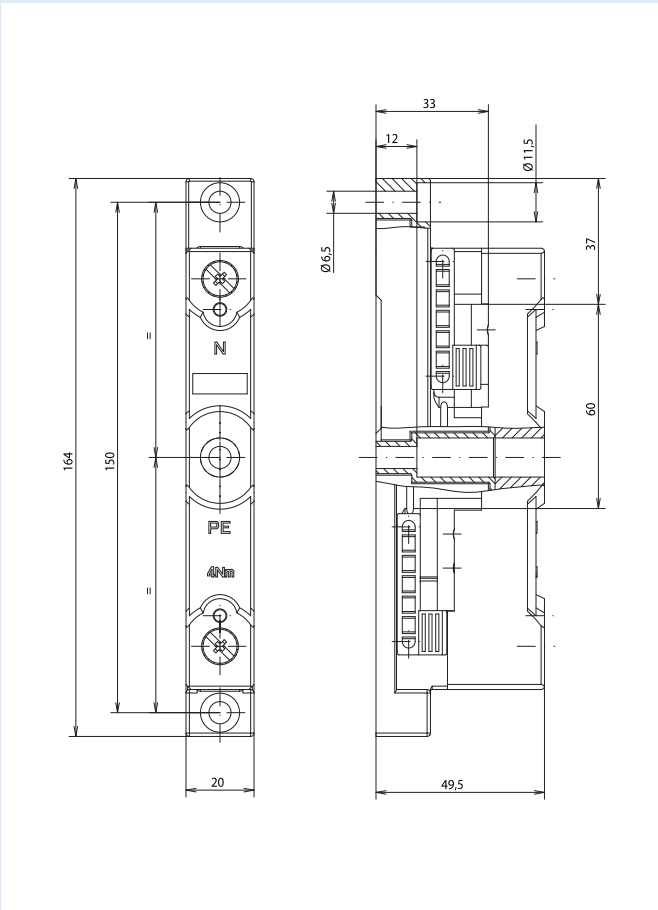
BBS-3/FL-NA



ES-BBS-3/FL



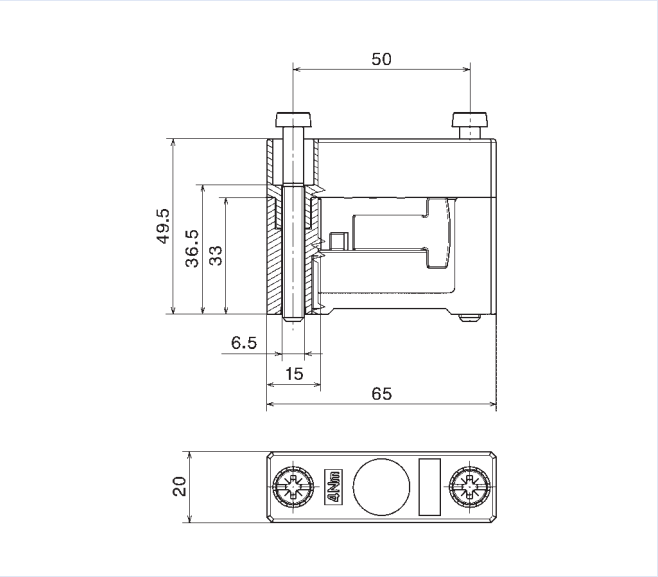
BBS-2/FL



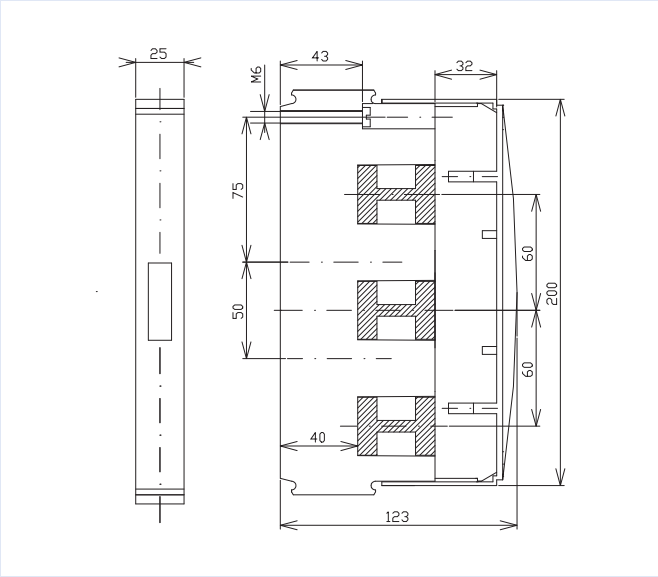
SASY 60i Busbar System

Dimensions

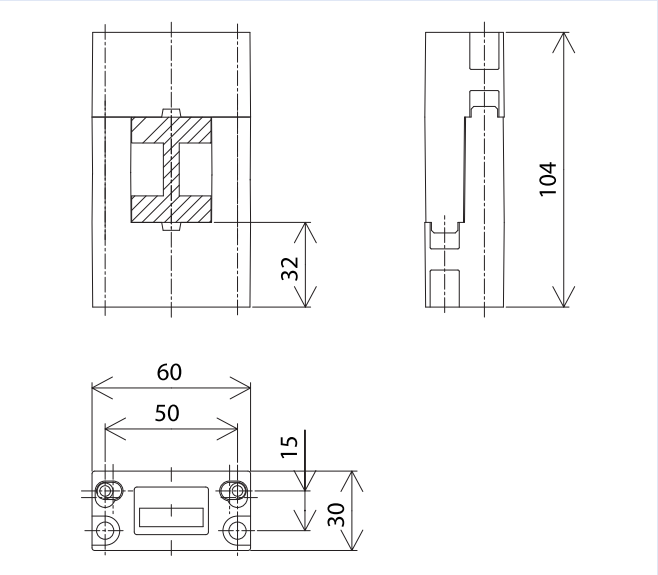
BBS-1/FL



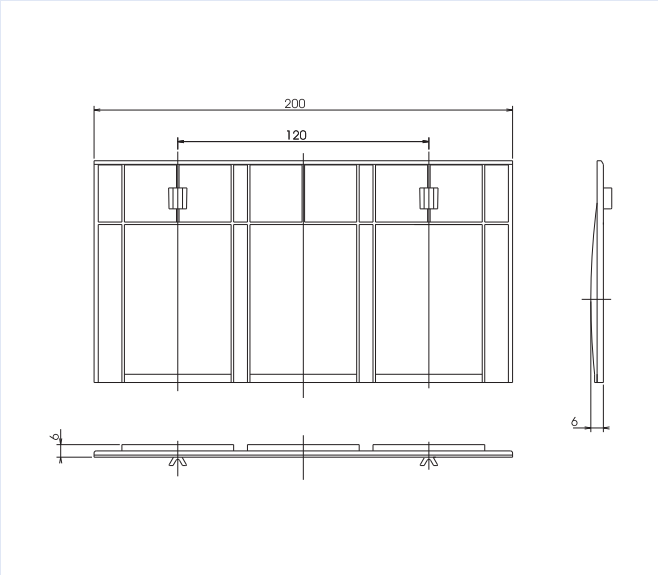
BBS-3/PR



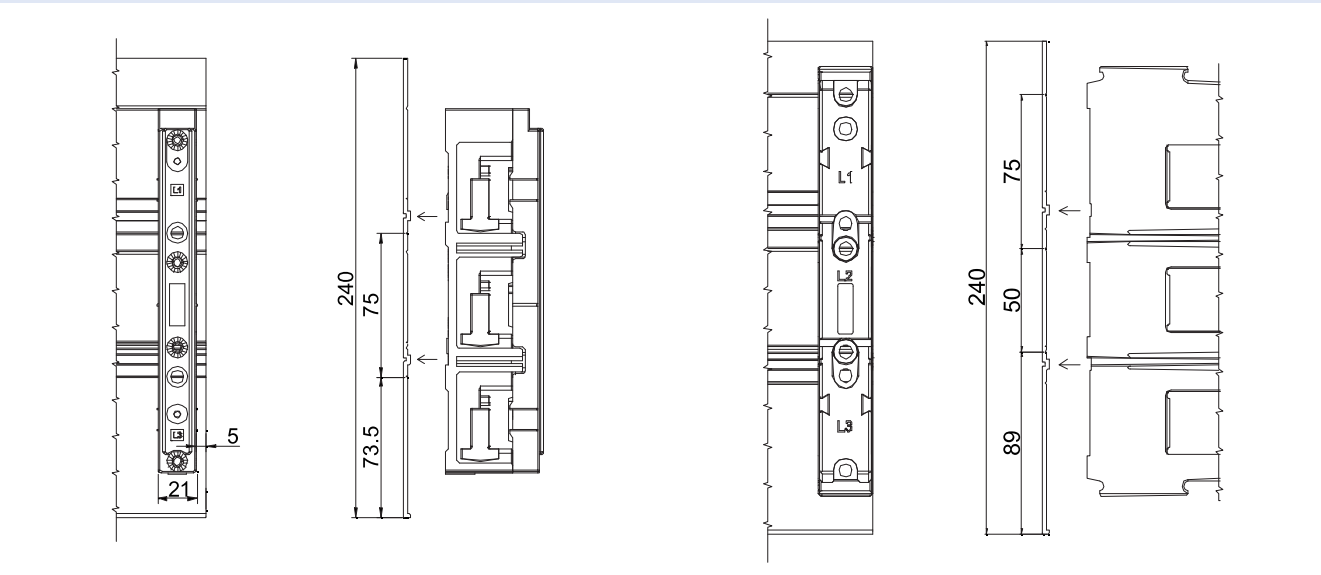
BBS-1/PR



ES-BBS-3/PR



BBC-BT-NA



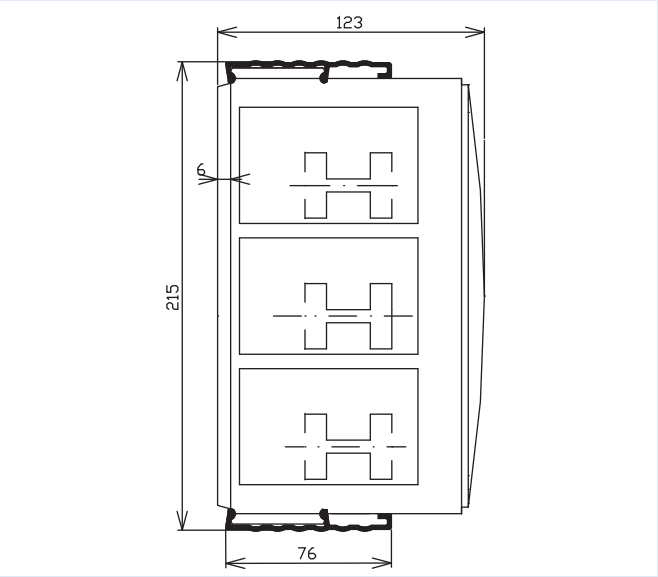
SASY 60i Busbar System

Dimensions	
CU-BAR-500/T	CU-BAR-720/T
BBC-CU-BAR/PR	BBC-RCOV1, BBC-MRCOV1
BBC-CS1, BBC-CS2	BBC-CS48/PR
a (mm) b (mm) c (mm) 228 194 214	

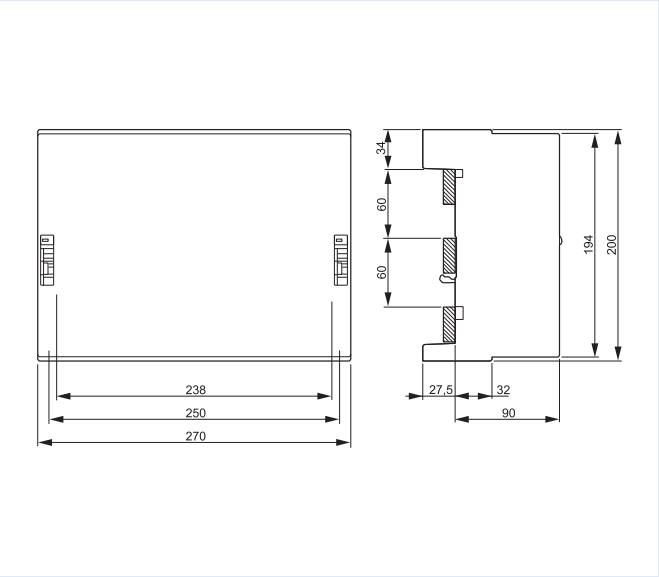
SASY 60i Busbar System

Dimensions

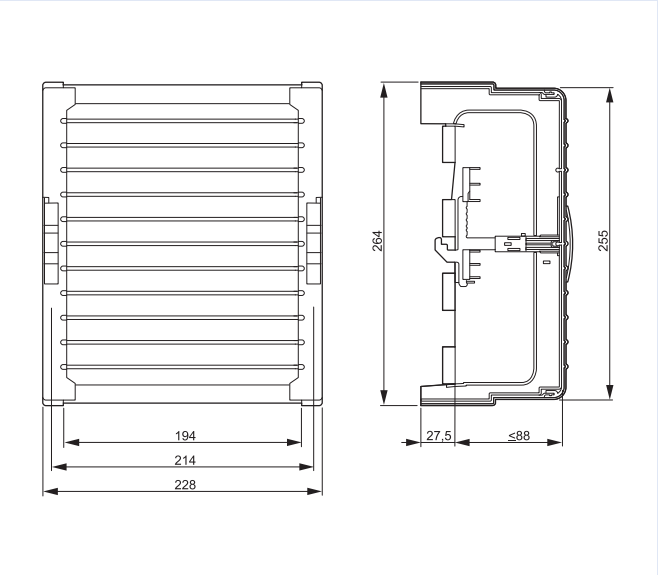
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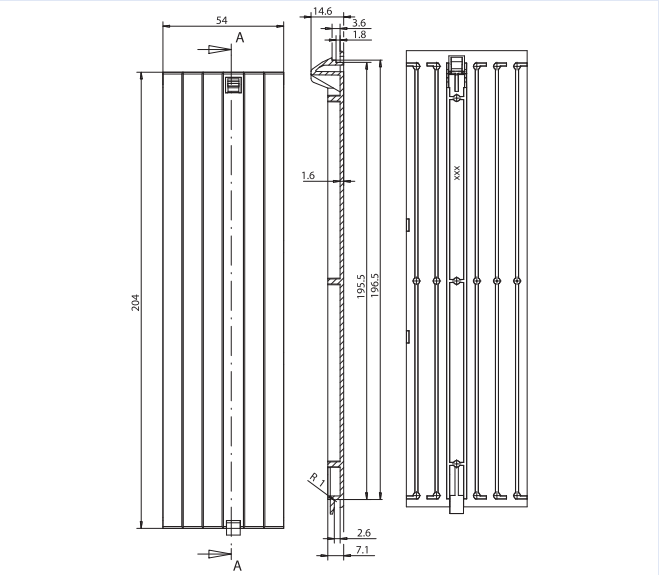
BBC-CS3



BBC-CS4



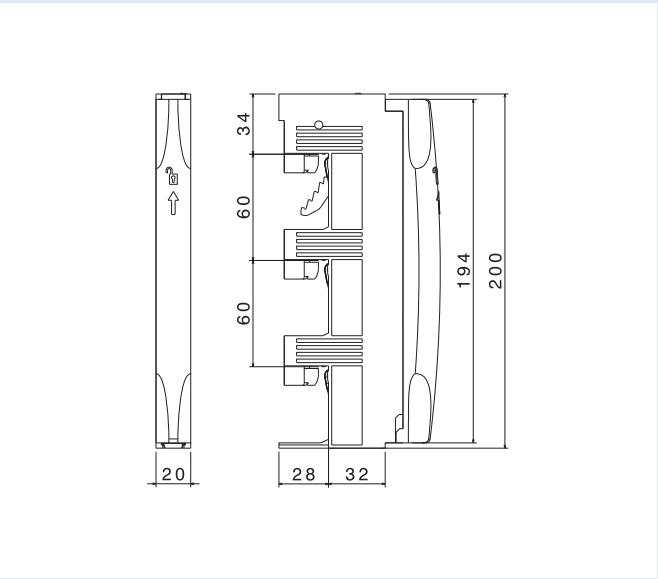
AM-195/54



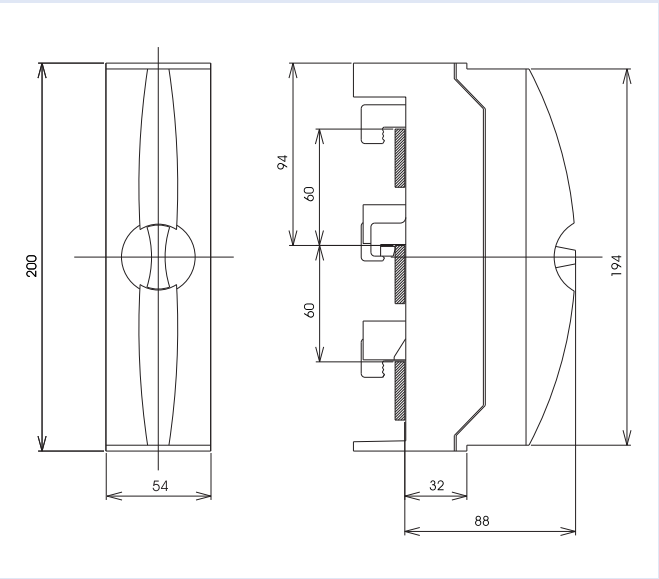
SASY 60i Busbar System

Dimensions

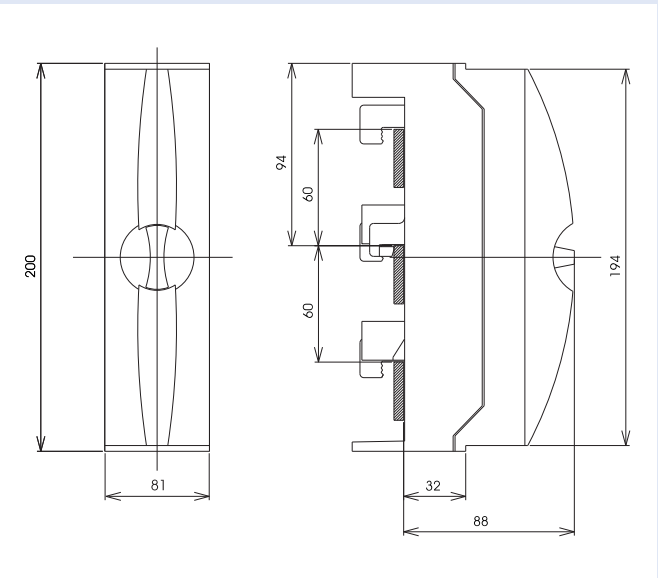
BBA-TP3/16



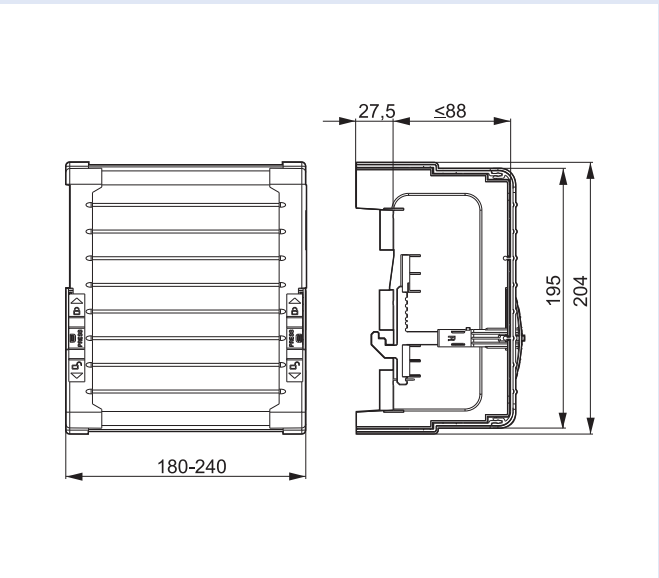
BBA-TP3/50



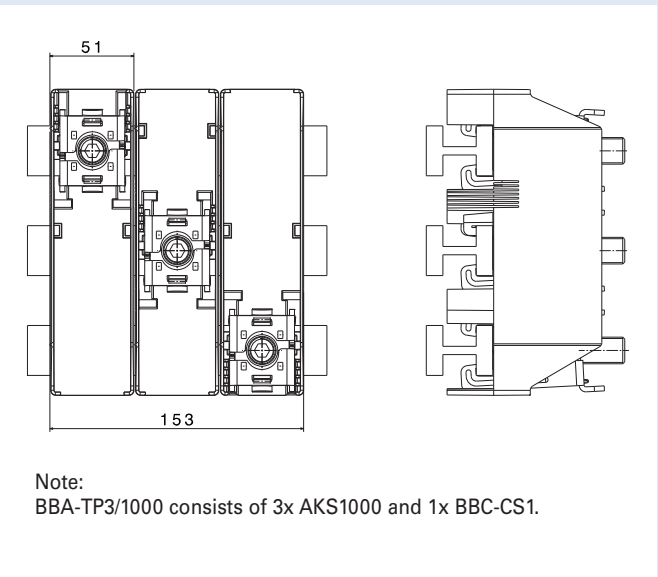
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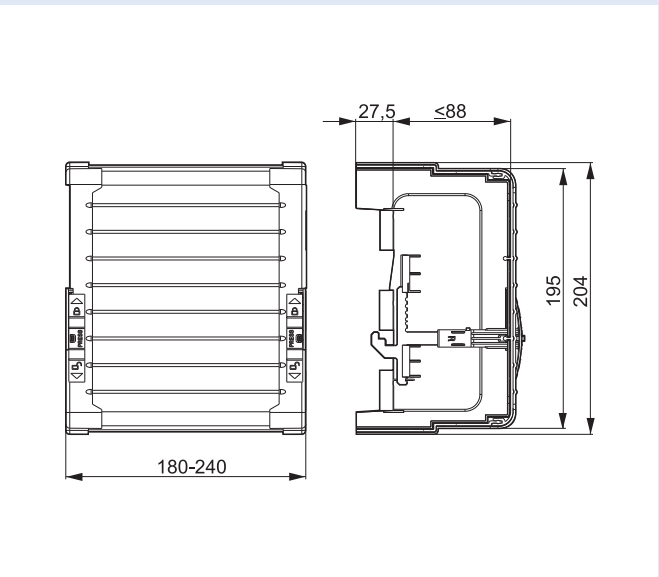
BBA-TP3/300



BBA-TP3/CU-BAND



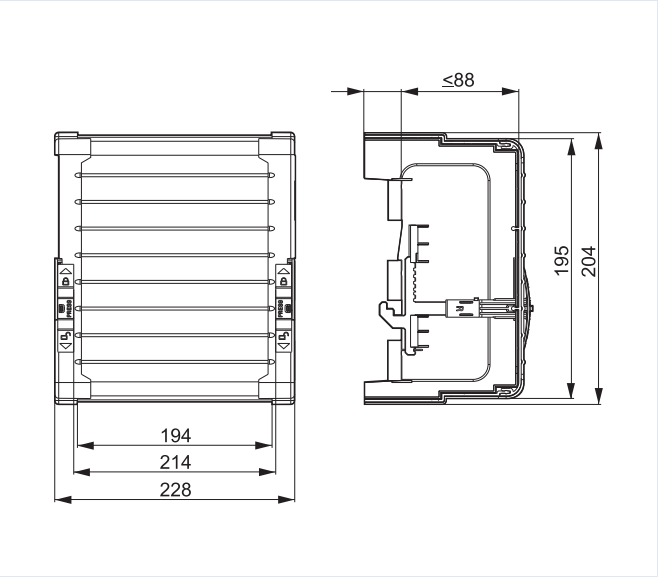
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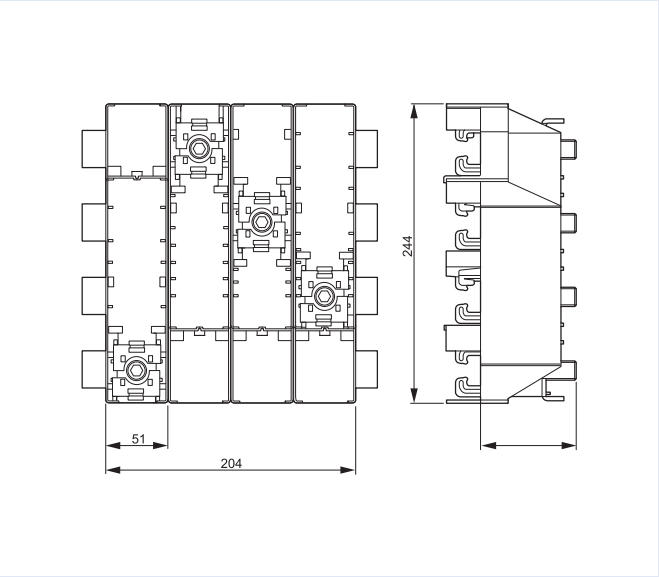
SASY 60i Busbar System

Dimensions

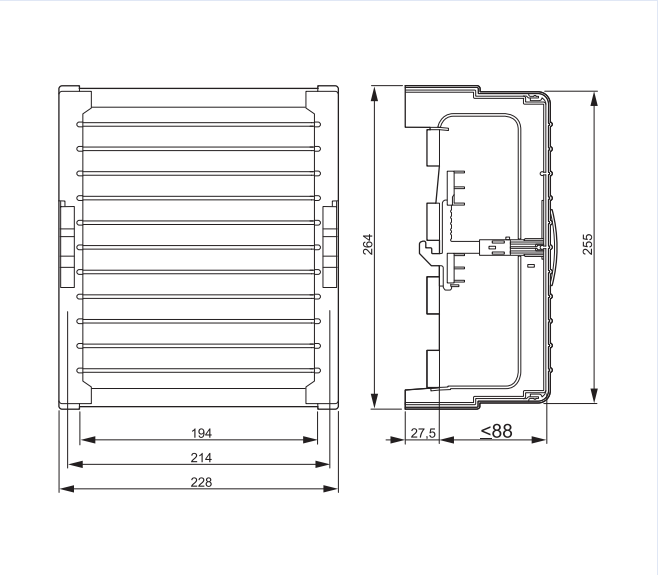
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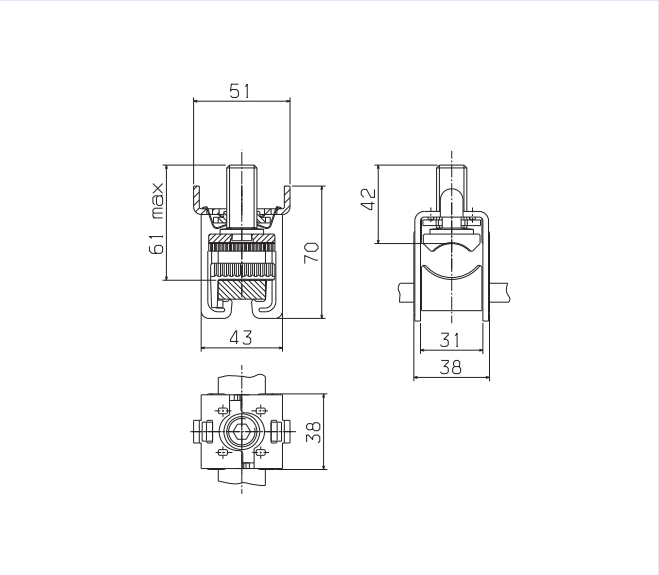
BBA-TP4/CU-BAND



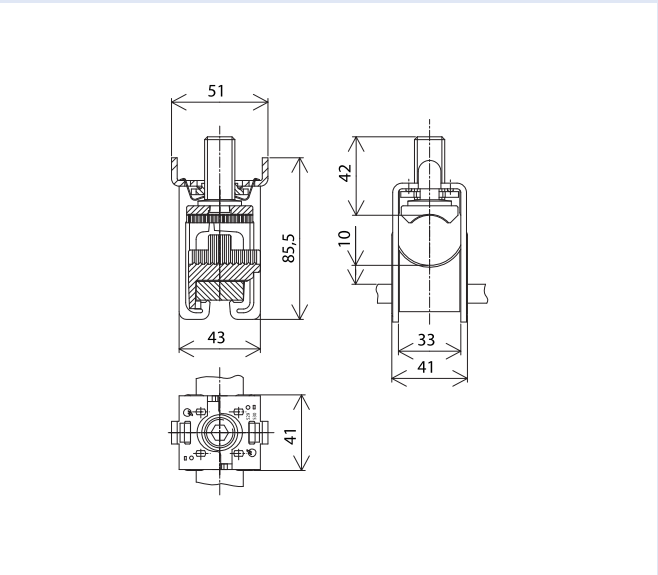
BBA-TP4/300



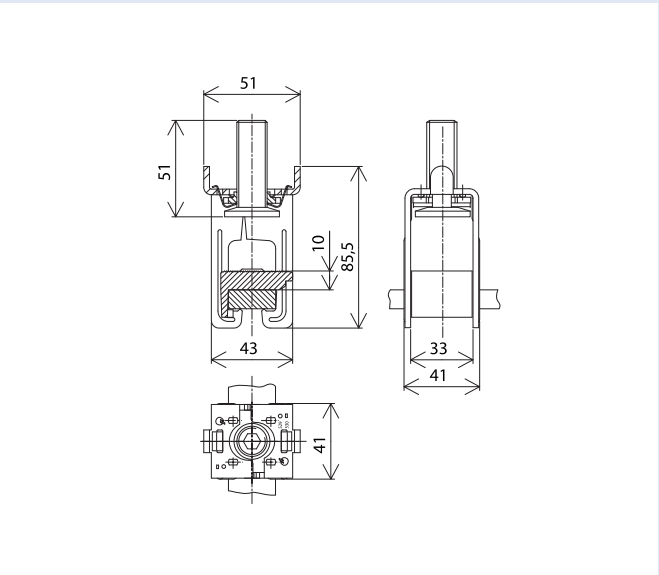
AKS185



AKS300



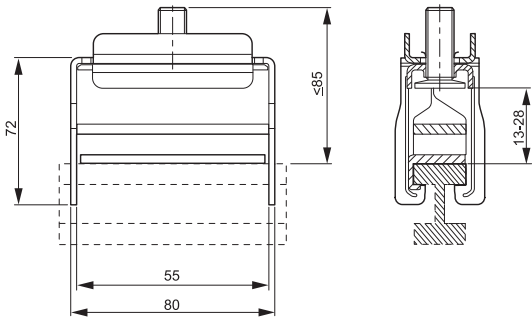
AKS-CU-BAND



SASY 60i Busbar System

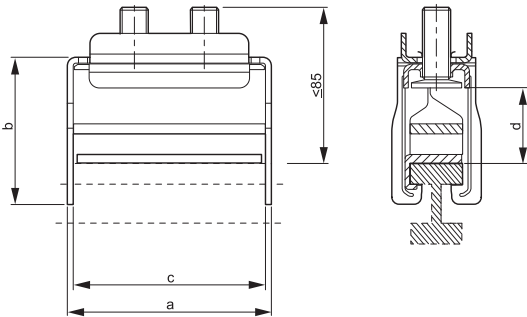
Dimensions

AKS1000



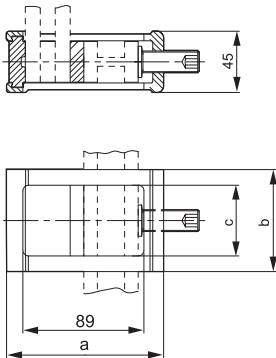
AKS1200, AKS2000

Type	a (mm)	b (mm)	c (mm)	d (mm)
AKS1200	85	80	68	13-38
AKS2000	122	80	105	13-38



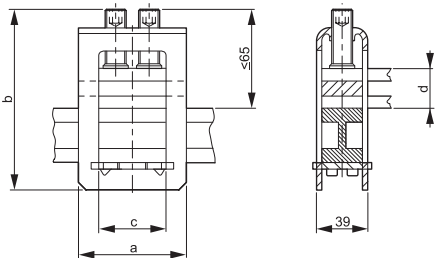
AKP800, AKP1000

Type	a (mm)	b (mm)	c (mm)
AKP800	118	72	41
AKP1000	103	94	64

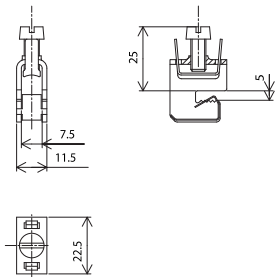


AKP750-AKP3600

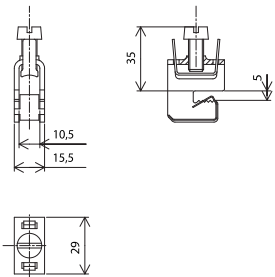
Type	a (mm)	b (mm)	c (mm)	d (mm)
AKP750	82	103	51	5-28
AKP900	94	103	64	5-28
AKP1200	94	118	64	20-42
AKP1600	112	118	81	20-42
AKP2000	132	118	101	20-42
AKP3600	132	154	101	23-45



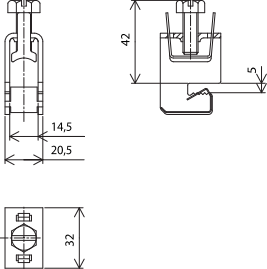
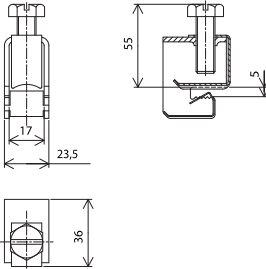
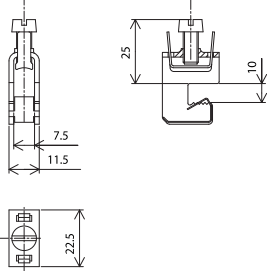
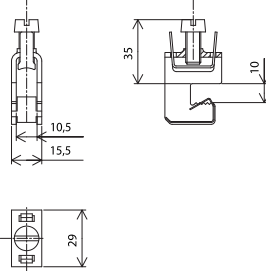
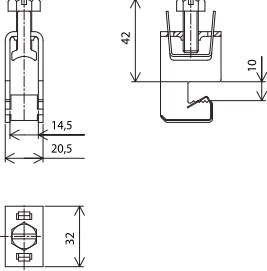
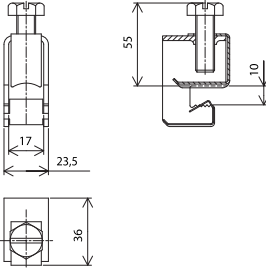
AKU 16/5



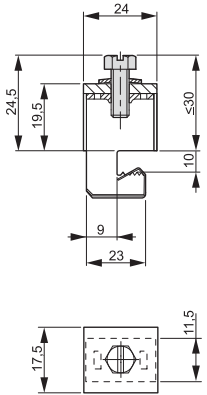
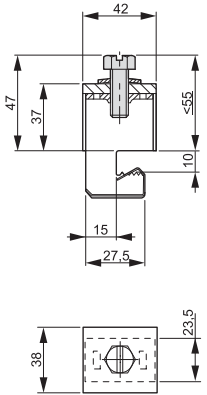
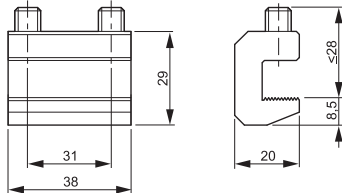
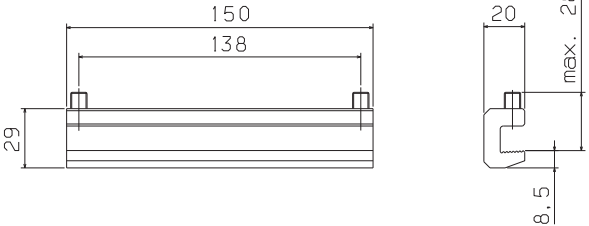
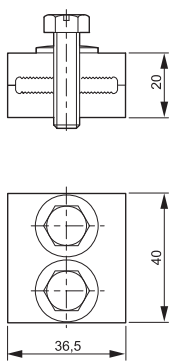
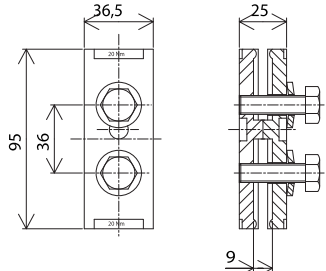
AKU 35/5



SASY 60i Busbar System

Dimensions	
AKU 70/5	AKU 120/5
	
AKU 16/10	AKU 35/10
	
AKU 70/10	AKU 120/10
	

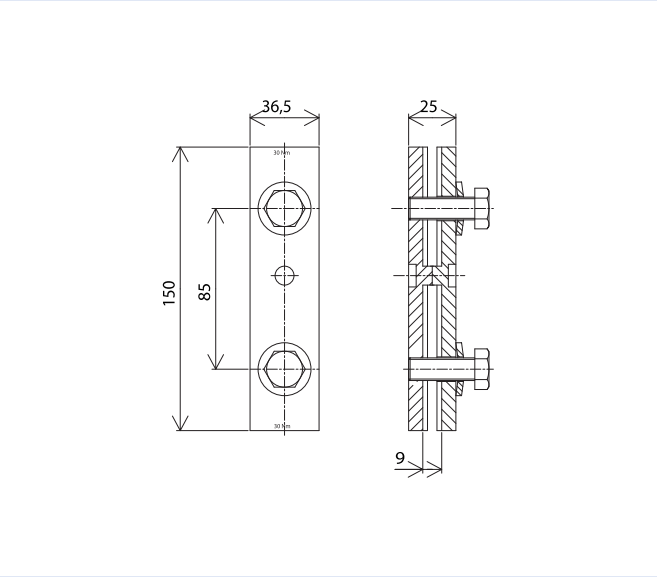
SASY 60i Busbar System

Dimensions	
AKUM8/10	AKUM10/10
	
BBT-CU12-20X5/10-38	BBT-CU12-20X5/10-150
	
BBT-CU20-30X5/10-40	BBT-CU20-30X5/10-95
	

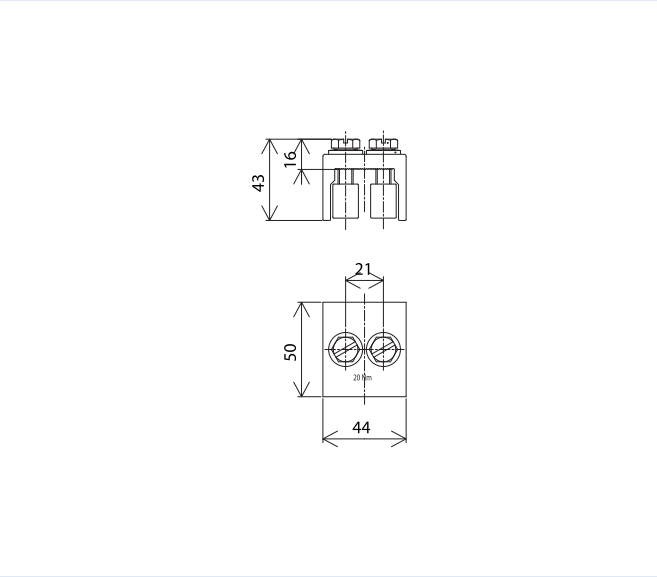
Sammelschienen-System SASY 60i

Dimensions

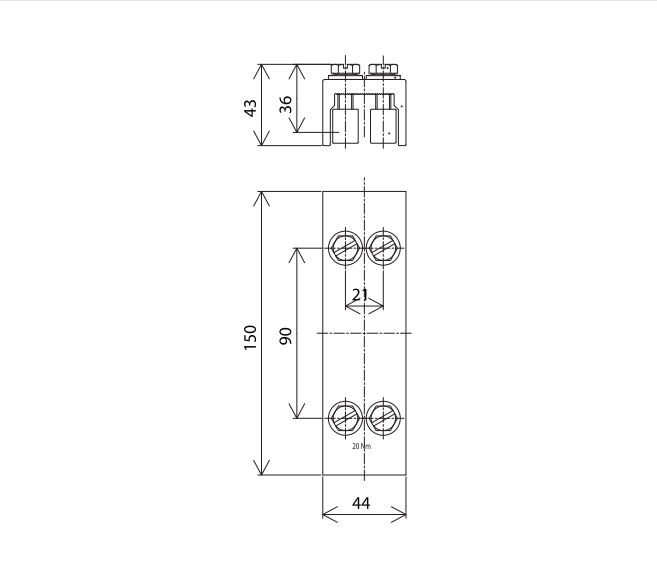
BBT-CU20-30X5/10-150



BBT-CU-BAR500/720-50



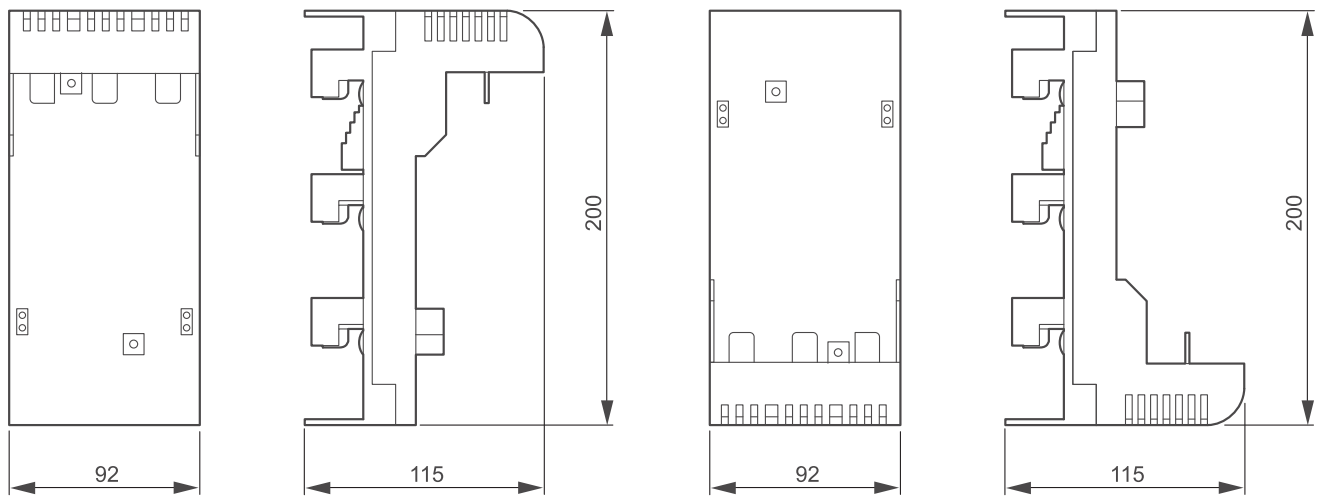
BBT-CU-BAR500/720-150



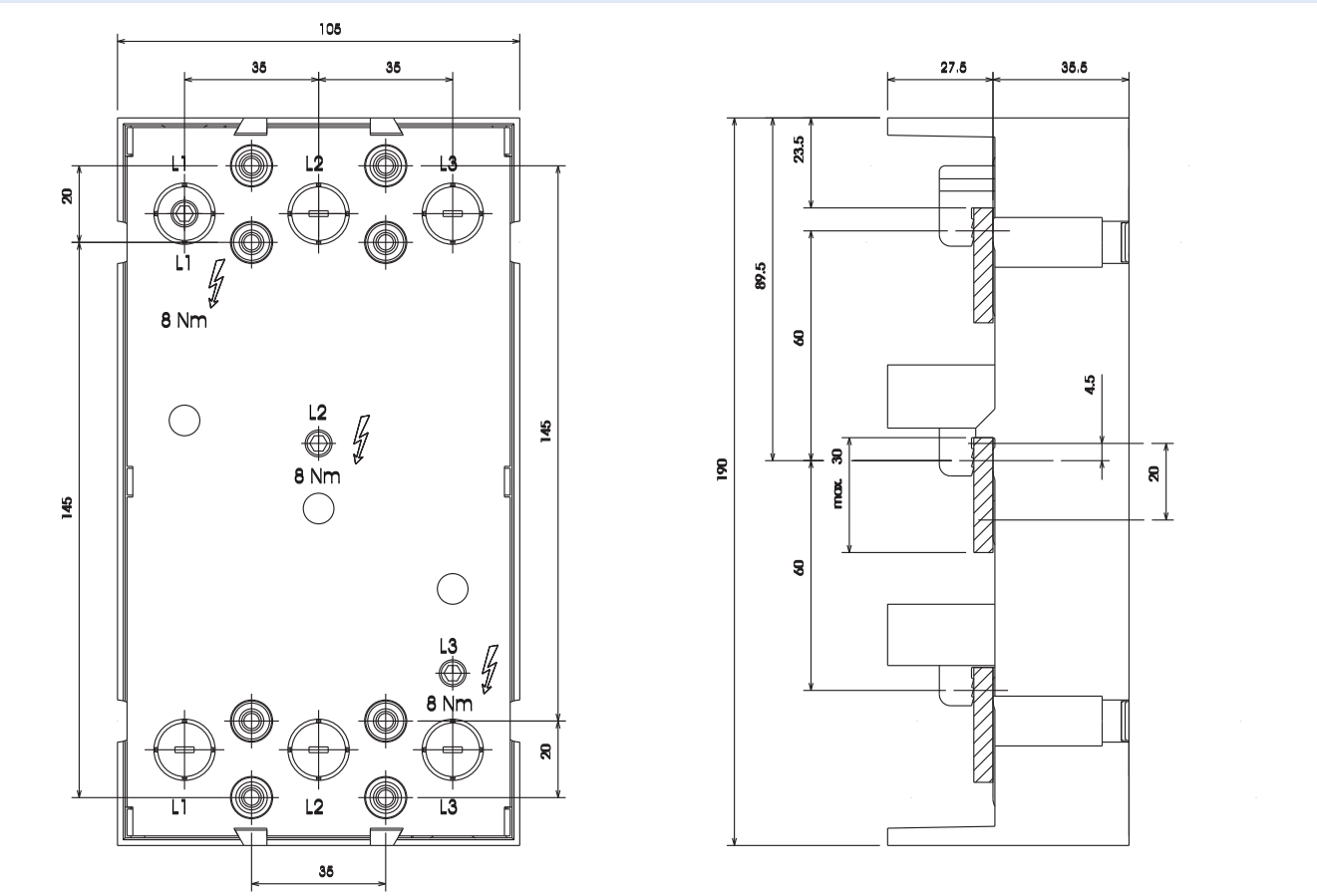
SASY 60i Busbar System

Dimensions

NZM1-XAD160



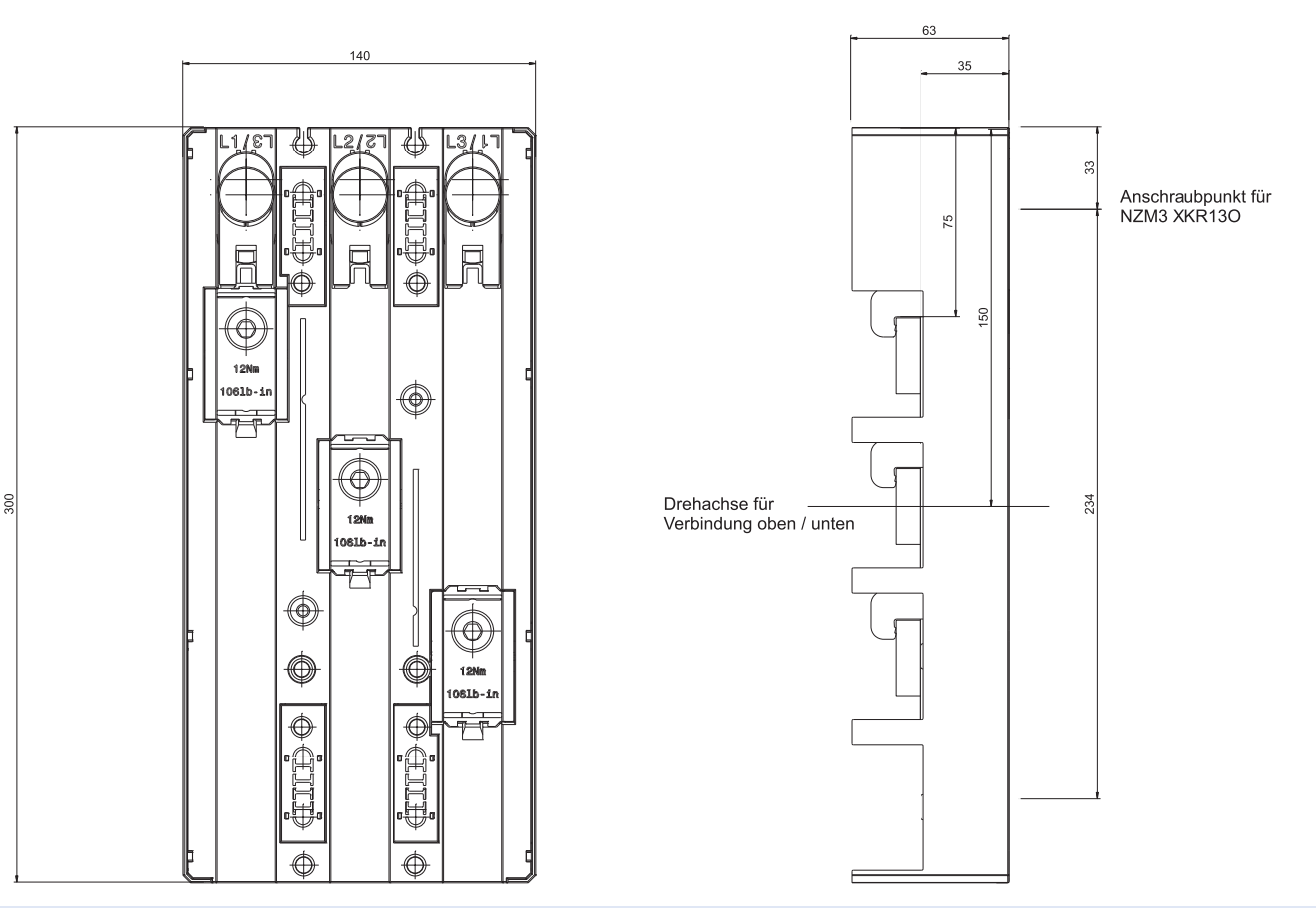
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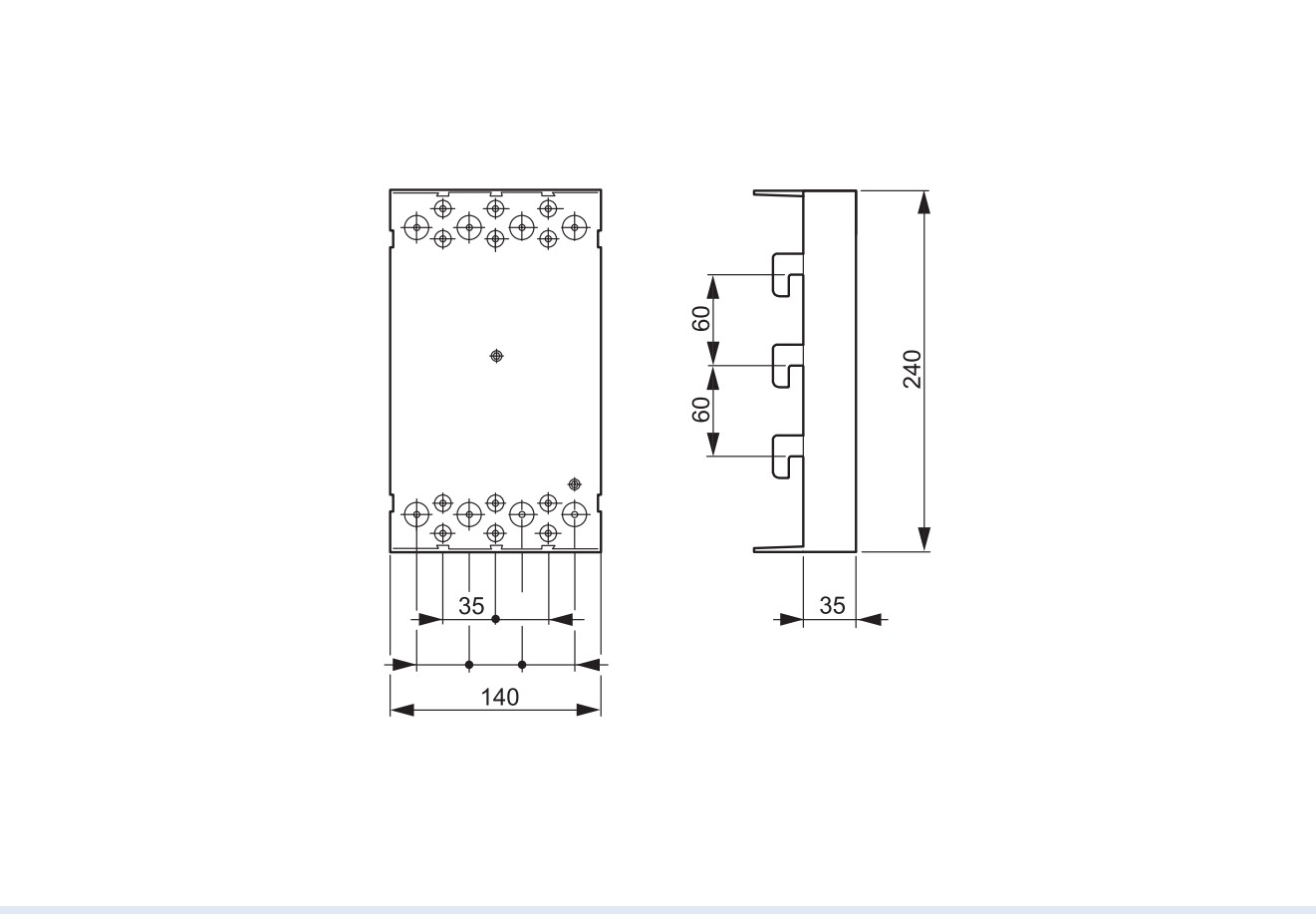
SASY 60i Busbar System

Dimensions

NZM3-XAD630



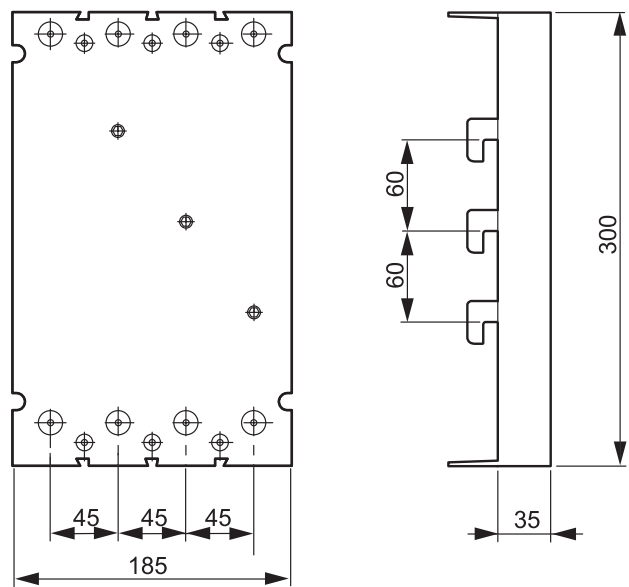
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SASY 60i Busbar System

Dimensions

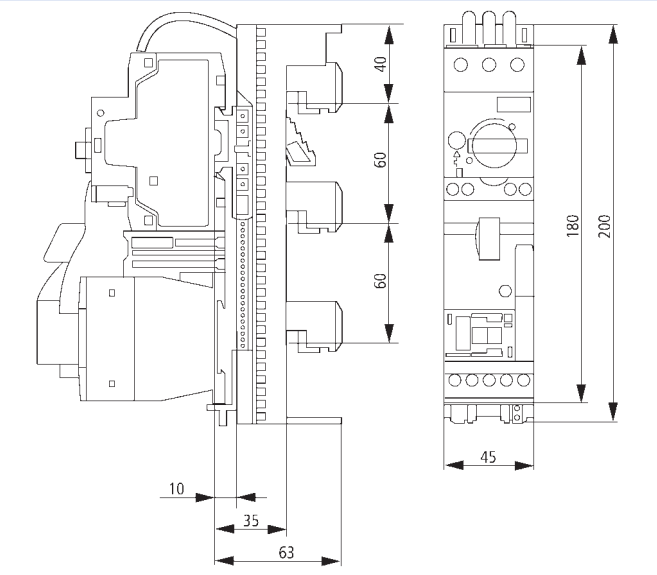
NZM3-4-XAD630



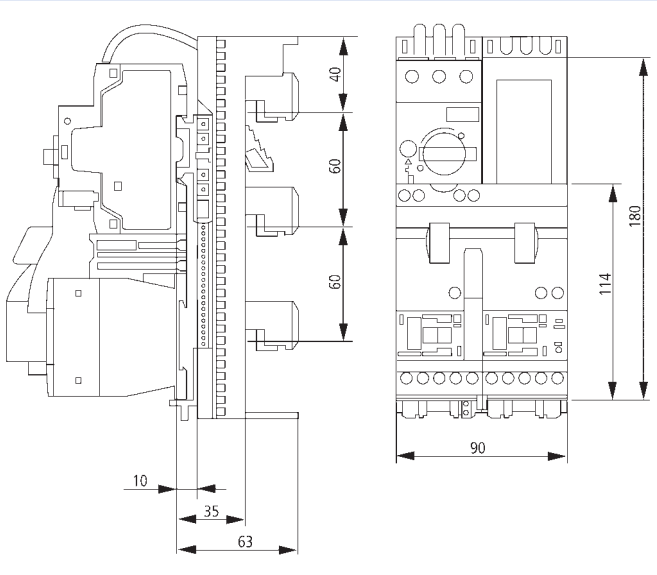
SASY 60i Busbar System

Dimensions

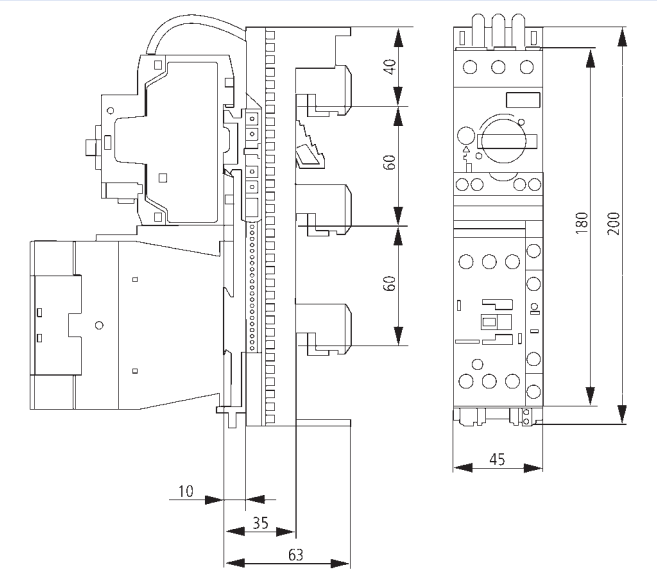
BBA0-25



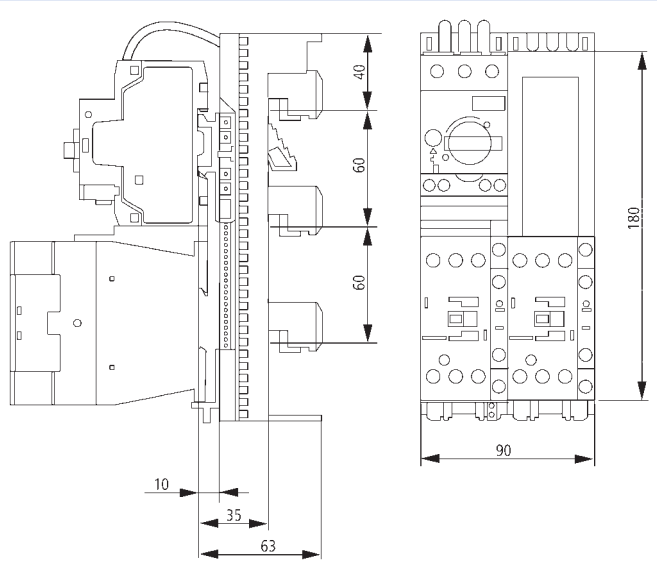
BBA0R-25



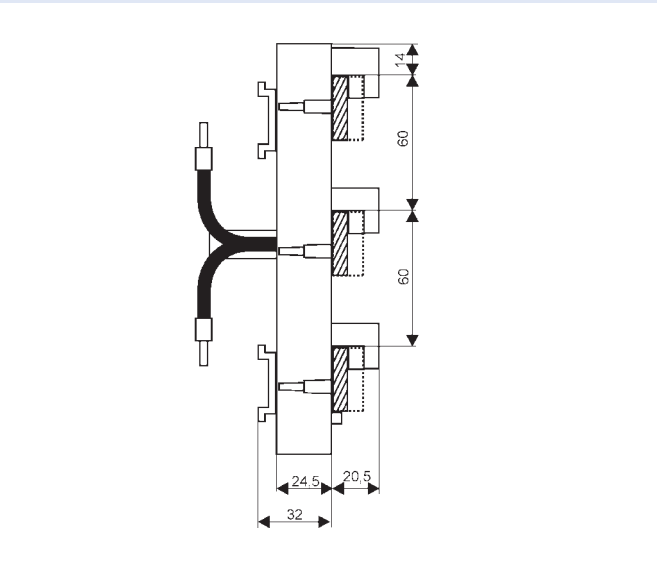
BBA0-32



BBA0R-32



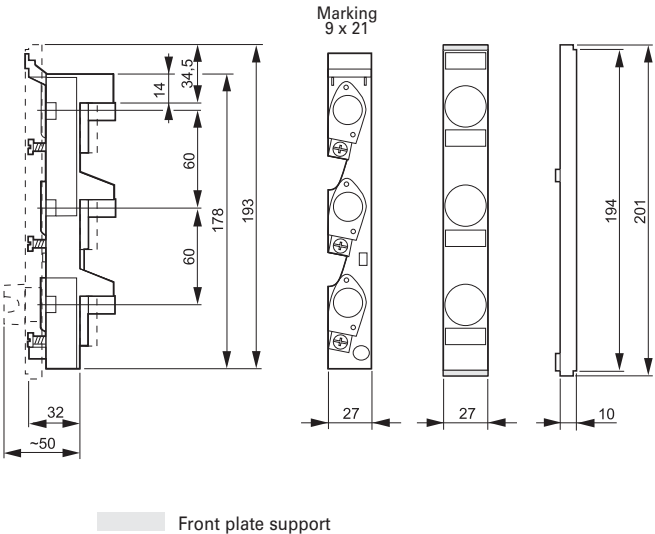
Z-SS-60-ADD/6...



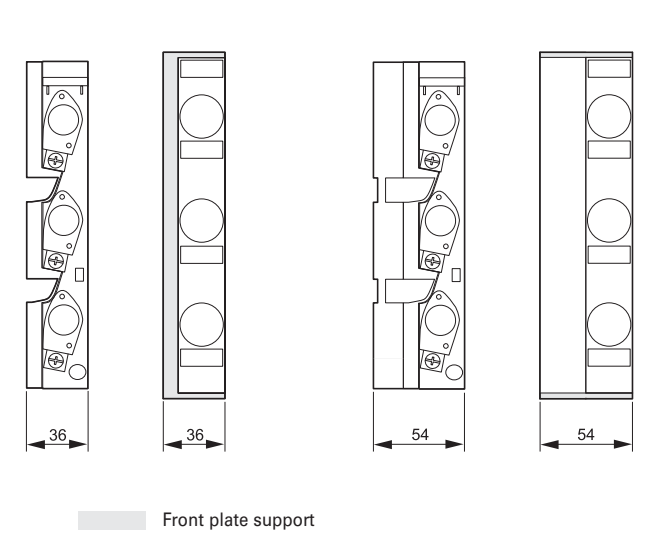
SASY 60i Busbar System

Dimensions

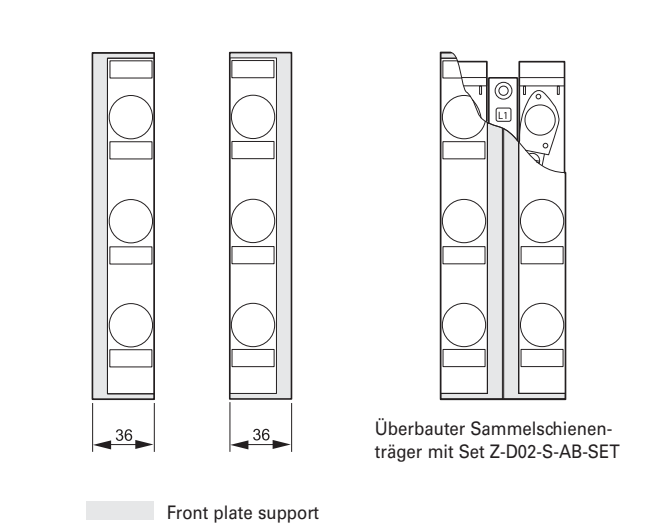
D02-SO/63/3-R-27



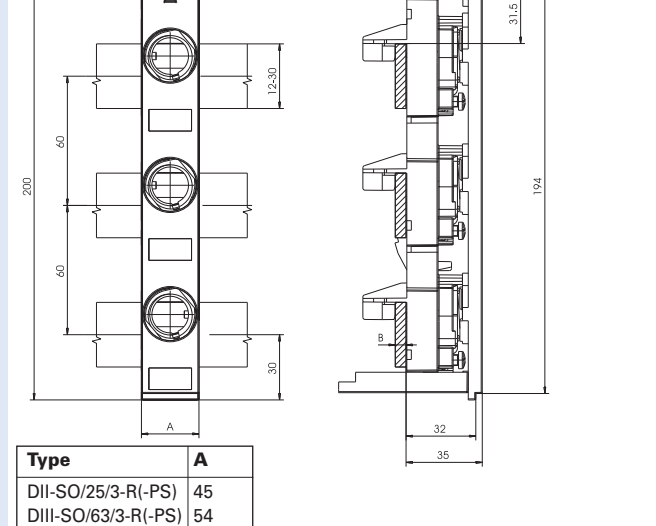
Z-D02/R/3-36, Z-D02/R/3-54



Z-D02-S-AB-SET

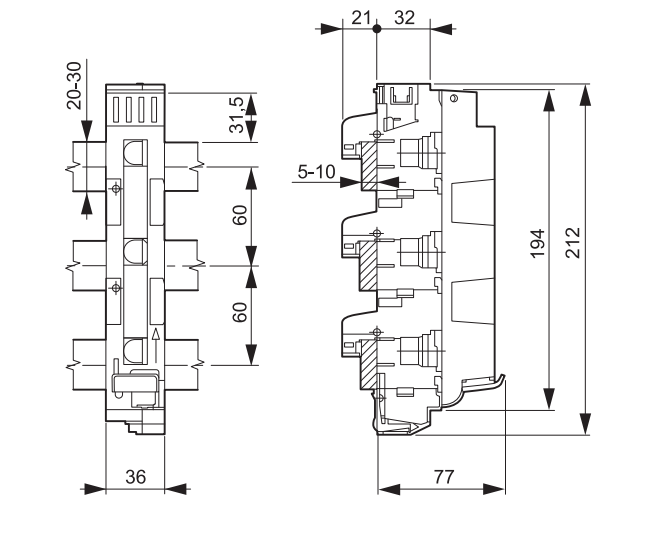


D...SO/.../3-R(-PS)

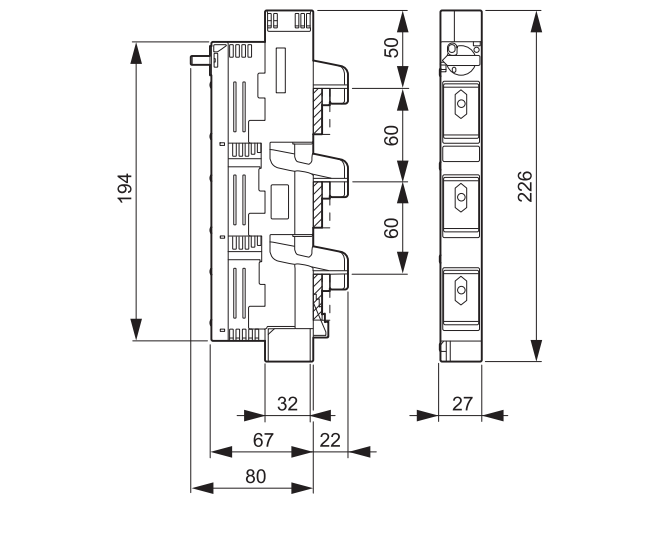


Type	A
DII-SO/25/3-R(-PS)	45
DIII-SO/63/3-R(-PS)	54

D02-S/63/3-RS



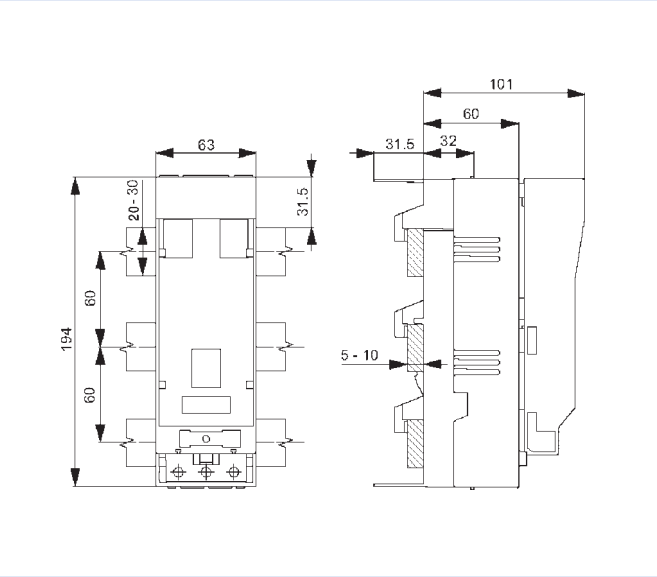
D02-LTS/63/3-R(-HK)



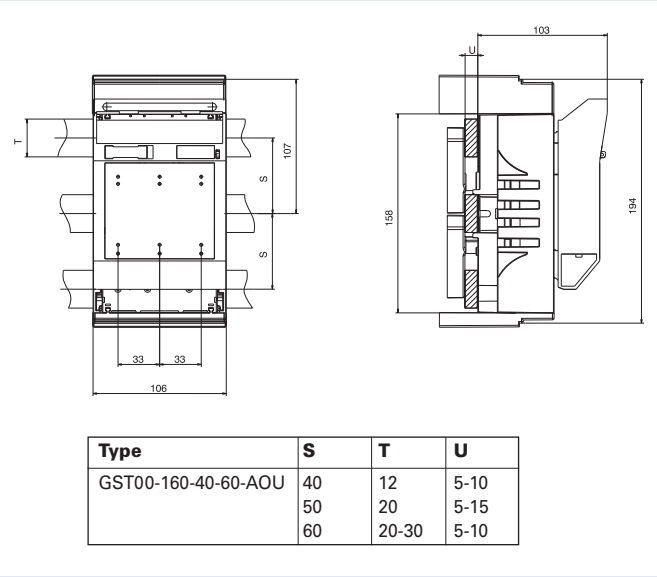
SASY 60i Busbar System

Dimensions

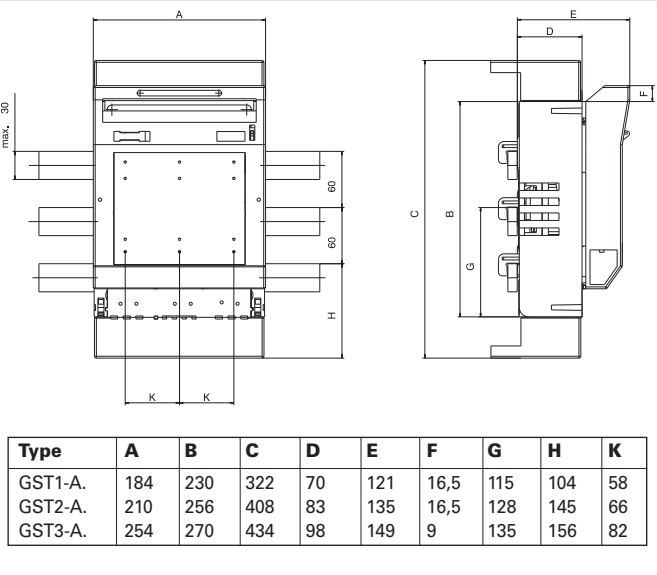
LTS-100/C00/3-R



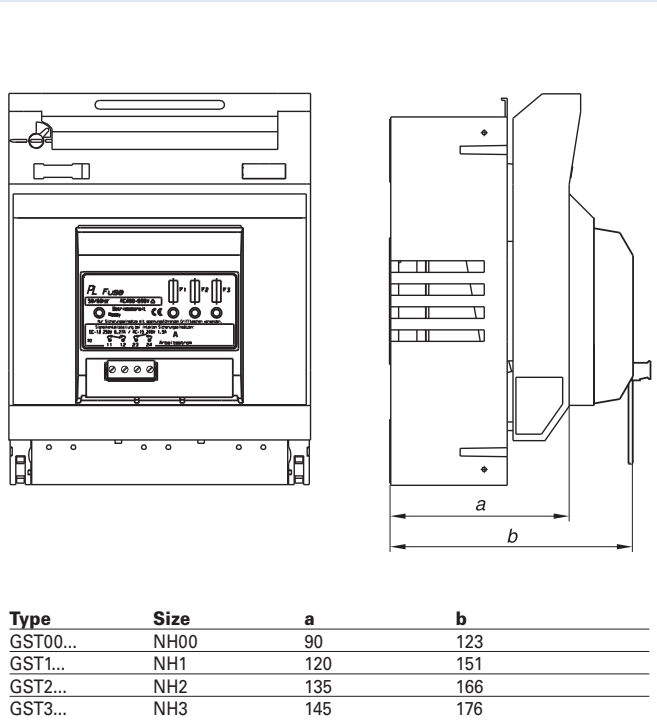
GST00-160-40-60-AOU



GST.-AO / GST.-AU



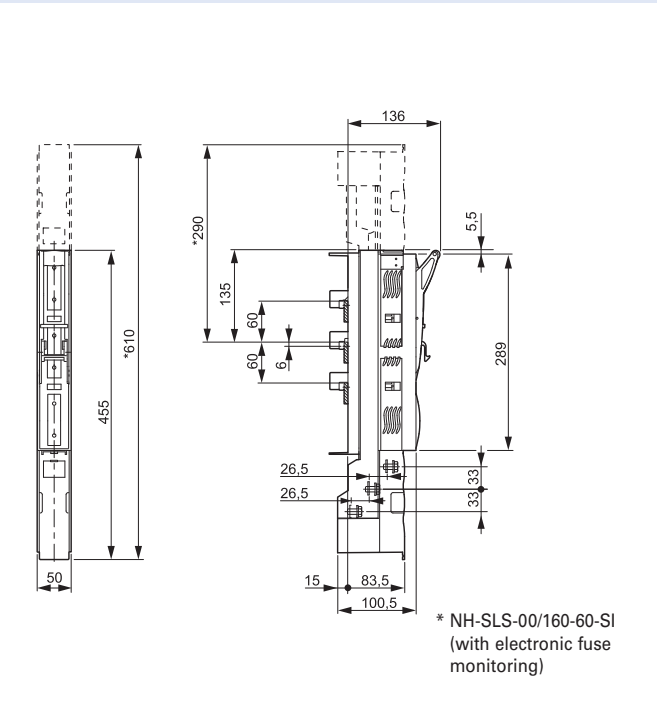
GST...-DSI



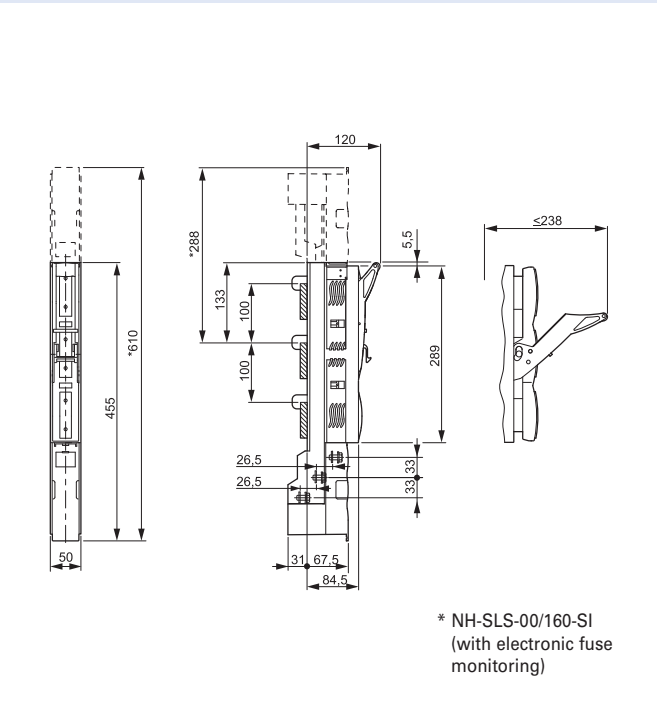
SASY 60i Busbar System

Dimensions

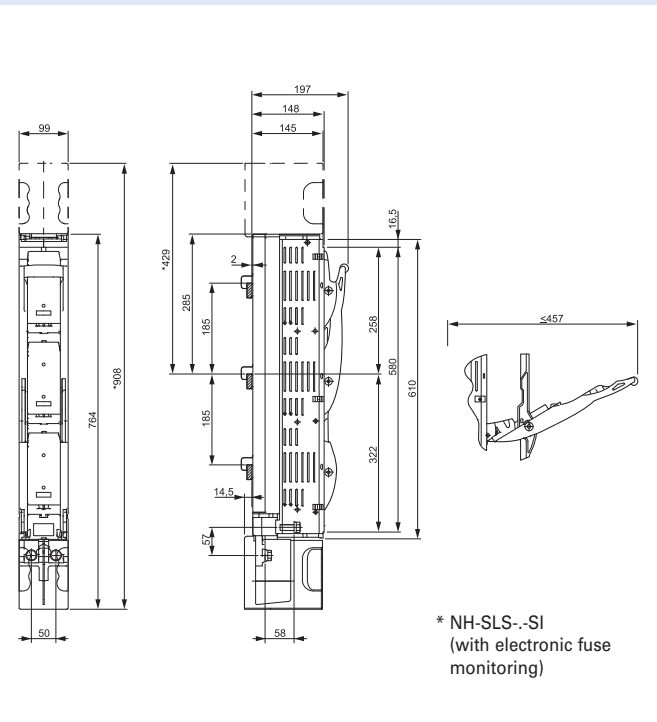
NH-SLS-00/160-60(-SI)



NH-SLS-00/160(-SI)

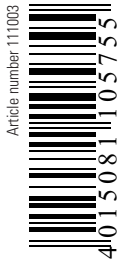


NH-SLS-1, NH-SLS-2, NH-SLS-3



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